

Team 22257 — Meeting Task Cards

Herobotix - Heritage Academy Gateway · 2026-2027 BIOBUZZ · REV Mecanum Drivetrain Kit v2 (REV-45-2470)

How this works

Every meeting starts with **10 minutes** of orientation together and ends with **10 minutes** of cleanup together. In between you pick a card and work it. **Every card is built to fill 100 minutes** — that is a whole Tuesday (4:00–6:00). Friday (12:00–3:00) gives you 160 minutes, so you get your card plus another hour: finish it properly, do the “If you finish early” box, or start the next card in your chain. Work the checklist top to bottom and get the “Done when” boxes checked before cleanup. If a card says you need another card first, do that one first.

ENGINEERING & PLANNING 7

EP-1 Team Identity: Name, Look &

Motto Start here

NO HARDWARE

EP-2 Read the Game, Build the Wall

Poster Start here

NO HARDWARE

EP-3 Robot Requirements List Some

experience

NO HARDWARE

needs EP-2

EP-4 Design Sketches & Pick a

Concept Some experience

NO HARDWARE

needs EP-3

EP-5 Set Up the Engineering Notebook Start

here

NO HARDWARE

EP-6 Tape Out the Practice Field Start here

NO HARDWARE

EP-7 Shop Safety & Tool Check-Off Start here

NO HARDWARE

CAD & 3D PRINTING 7

CD-1 Onshape Setup & Your First Part Start

here

NO HARDWARE

CD-2 Measure & Model a Real Robot

Part Some experience

NO HARDWARE

needs CD-1

CD-3 Design a Pollen Intake in

CAD Challenge

NO HARDWARE

needs CD-2, EP-4

CD-4 Design a Pollen Outtake in

CAD Challenge

NO HARDWARE

needs CD-2, EP-2

CD-5 Build the Robot in CAD: Assembly &

Fit Check Challenge

NO HARDWARE

needs CD-2

CD-6 From CAD to Printed Part Some

experience

NO HARDWARE

needs CD-2

CD-7 Design a Printed Part That Solves a

Real Problem Challenge

NO HARDWARE

needs CD-6

PROGRAMMING 14

PR-1 Get Connected: Control Hub & Driver Station [Start here](#)

BOARD OR ROBOT

PR-2 Build the Robot Configuration File [Start here](#)

BOARD OR ROBOT

needs PR-1

PR-3 First TeleOp: Make It Drive (Mecanum) [Start here](#)

ROBOT

needs PR-2

PR-4 Telemetry: See What the Robot Is Thinking [Some experience](#)

BOARD OR ROBOT

needs PR-3

PR-5 Driver Tuning: Slow Mode & Smooth Control [Some experience](#)

ROBOT + FIELD

needs PR-3, EP-6

PR-6 First Autonomous: Time-Based Movement [Some experience](#)

ROBOT + FIELD

needs PR-3, EP-6

PR-7 Encoder Autonomous: Drive an Exact Distance [Challenge](#)

ROBOT + FIELD

needs PR-6, PR-4

PR-8 Intake Control: Map a Mechanism to a Button [Some experience](#)

ROBOT

needs PR-2, BA-5

PR-9 Practice Without the Robot: FTC SIM [Start here](#)

NO HARDWARE

PR-10 Touch Sensor: The First Decision [Start here](#)

PROGRAMMING BOARD

needs PR-2

PR-11 Color Sensor: Teach It to See Pollen [Some experience](#)

PROGRAMMING BOARD

needs PR-4

PR-12 Potentiometer & Servo: Analog In, Position Out [Some experience](#)

PROGRAMMING BOARD

needs PR-4

PR-13 Bench to Robot: Move Your Code Over [Challenge](#)

ROBOT

needs PR-3, PR-10

PR-14 Field-Centric Drive with the IMU [Challenge](#)

ROBOT + FIELD

needs PR-3, PR-4

BUILDING & ASSEMBLY 9

BA-1 Inventory the Kit & Set Up the Shop [Start here](#)

NO HARDWARE

BA-2 Mecanum Chassis: Build the Frame [Start here](#)

ROBOT

needs BA-1, EP-7

BA-3 Mecanum Drive: Gearboxes, Motors & Wheels [Some experience](#)

ROBOT

needs BA-2

BA-4 Electronics: Mount & Wire the Control Hub [Some experience](#)

ROBOT

needs BA-3, EP-7

BA-5 Design & Build an Intake [Challenge](#)

ROBOT

needs BA-3, EP-4

BA-6 Battery Management Station [Start here](#)

NO HARDWARE

BA-7 Scoring Mechanism Prototype (cardboard first) [Challenge](#)

ROBOT

needs BA-5, EP-4

BA-8 Robot Cart & Pit Kit [Start here](#)

ROBOT

BA-9 Assemble a Programming Board [Start here](#)

PROGRAMMING BOARD

needs EP-7

MERCH & GIVE-AWAY 6

MG-1 Give-Away Plan & Budget Start here

NO HARDWARE

needs EP-1

MG-2 Design the Give-Away Art Some experience

NO HARDWARE

needs EP-1, MG-1

MG-3 Produce the Give-Aways (assembly line) Start here

NO HARDWARE

needs MG-2

MG-4 Team Card / Business Card Some experience

NO HARDWARE

needs EP-1

MG-5 3D-Printed Give-Away Challenge

NO HARDWARE

needs MG-1

MG-6 Design the Award Patch Set Some experience

NO HARDWARE

needs EP-1

TEAM T-SHIRTS 4

TS-1 Shirt Design Concepts Start here

NO HARDWARE

needs EP-1

TS-2 Vote & Make the Art Print-Ready Some experience

NO HARDWARE

needs TS-1

TS-3 Sizing Survey & Order Sheet Start here

NO HARDWARE

TS-4 Get Quotes & Recommend a

Printer Some experience

NO HARDWARE

needs TS-2, TS-3

MARKETING & FUNDRAISING 7

MF-1 Build the Sponsor Prospect List Start here

NO HARDWARE

MF-2 Sponsorship One-Pager Some experience

NO HARDWARE

needs EP-1, MF-1

MF-3 Write the Outreach Emails & Letters Some experience

NO HARDWARE

needs MF-2

MF-4 Social Media Plan & First Posts Start here

NO HARDWARE

needs EP-1

MF-5 Team Page Content for the Herobotix Website Some experience

NO HARDWARE

needs EP-1

MF-6 Plan a Demo / Outreach Event Some experience

ROBOT

needs PR-3

MF-7 Grants & Team Funding

Applications Challenge

NO HARDWARE

needs MF-2

CHOOSING CARDS — THE RULES THAT KEEP THIS WORKING

1. **Read “Before you start” first.** If a card names another card, that one has to be finished by somebody first. Most cards can be started cold.
2. **Check the hardware tag.** It decides what can run at the same time. Only the robot cards compete for the robot; the programming board and the “no hardware” cards run alongside them.
3. **Every merch and t-shirt card except TS-3 waits on EP-1.** There is no team name until meeting 1 is done, so there is nothing to put on a shirt.
4. **Match the crew size on the card.** A card marked 2–3 does not absorb six people; it absorbs three and bores the other three. Run a second card instead.
5. **Rotate.** Nobody spends the whole season on one category. The point of a menu is that everyone tries things they would not have picked.
6. **A card is not done because time ran out.** It is done when the “Done when” boxes are true. If it did not finish, note where it stopped and hand the card to the next crew.
7. **Anything leaving the team needs coach approval** — emails to businesses, social posts, orders, anything with a student’s name or face on it.

RESOURCE SHELF

Skill Builders online training (FIRST Dashboard login)

<https://training.firstinspires.org/learn/course/ftc-skill-builders-26-27>

REV Mecanum Drivetrain V2 build guide - the student robot

<https://docs.revrobotics.com/duo-build/mecanum-drivetrain-v2>

REV UltraPlanetary gearbox docs

<https://docs.revrobotics.com/rev-crossover-products/ultraplanetary/up-gearbox>

REV Ultra 90 gearbox docs

<https://docs.revrobotics.com/rev-crossover-products/ultraplanetary/ultra-90>

REV StarterBot guide - reference only, not the team's build

<https://ftc-resources.firstinspires.org/ftc/archive/2027/team/rev-starterbot>

FTC Docs - programming reference

https://ftc-docs.firstinspires.org/en/latest/programming_resources/index.html

Game Manual Zero (gm0) - start here

<https://gm0.org/en/latest/docs/start-here.html>

FTC SIM - browser robot simulator

<https://ftcsim.org/>

Virtual Robot Simulator

<https://www.vrobotsim.online/homepage.html>

FTC Community Forum

<https://ftc-community.firstinspires.org/>

Onshape for FIRST teams (free CAD)

<https://www.onshape.com/en/education/first-robotics#ftc>

Learn Java for FTC - free PDF, optional reference

<https://github.com/alan412/LearnJavaForFTC>

Programming board 3D model - the board this team built

<https://www.thingiverse.com/thing:6864893>

Running the meeting

Coach page — Team 22257 · Tue 4:00–6:00 PM & Fri 12:00–3:00 PM, Room 200

Two meeting shapes

DAY	TASK TIME	HOW IT RUNS
Tuesday 4:00–6:00	100 min	10 min orientation → one card per crew → 10 min cleanup. A card is built to fill exactly this.
Friday 12:00–3:00	160 min	10 min orientation → one card, then 60 more minutes → 10 min cleanup. Spend the extra hour finishing, on the card's "If you finish early" box, or starting the next card in that crew's chain. On Fridays late in the season, use it to run a Skill Builders challenge as a whole team.

The first four meetings — then two weeks of fall break

The goal: a drivable robot base attempting the Fast Lane Challenge on Friday Sept 25, the last meeting before break. The student robot is the **REV Mecanum Drivetrain Kit v2** — drivetrain only, no intake needed for Fast Lane. That is four meetings to go from an unopened kit to a robot driving a taped track, so the build has a critical path and it is marked ► **CRITICAL** below. Everything else can slip; those cannot.

Coach checklist before Sept 15. Three things have lead time and will silently sink the deadline: **(1)** confirm a Control Hub is free for the student robot — the mecanum kit does not include one; **(2)** confirm the Mecanum Drivetrain Kit v2, a 12V battery, a switch and motor cabling are all on hand; **(3)** charge every battery. Order anything missing this week, not in two weeks.

MEETING	TASK TIME	CREWS AND CARDS
1 Tue Sep 15	100 min	Whole team, no crews yet. Run the short version of two cards: EP-1 Team Identity through the vote (~55 min — the team gets a name today), then ► EP-7 Safety & Tool Check-Off, safety walk plus hand tools only (~45 min). <i>EP-7 is on the critical path: nobody may touch the build on Friday without it.</i>
2 Fri Sep 18	160 min	Crews form — students pick a track. ► Build: BA-1 Inventory (compress to ~60) → BA-2 Drivetrain Frame (100). <i>BA-2 must finish today.</i> Programming: PR-1 Get Connected <i>on the dev board</i> (100) → start PR-2 Configuration (60). Media: EP-2 Read the Game (100) → start TS-1 Shirt Concepts (60). CAD: CD-1 Onshape Setup & First Part (100) → start CD-2 Measure & Model (60). <i>Needs no hardware and no robot — a fifth crew can run this from day one.</i>
3 Tue Sep 22	100 min	► Build: BA-3 Mecanum Drive — gearboxes, motors, wheels. <i>Must finish today.</i> Check the roller X pattern before they leave — four wheels in the wrong corners is the classic mecanum failure and it looks fine until the code will not strafe. Programming: finish PR-2 on the board, then PR-10 Touch Sensor. Send one programmer to shadow the build and write down the motor ports — they will configure the robot on Friday. ► Field: EP-6 Tape the Practice Field. <i>No field, no Fast Lane.</i>
4 Fri Sep 25 FAST LANE	160 min	0–100: ► BA-4 Wire the Control Hub (build crew) while the programming crew preps the robot's motor names and reviews what they learned on the board. 100–135: PR-2 on the <i>robot</i> — fast, because they already did this on the dev board — then PR-3 First TeleOp. On mecanum, PR-3 says prove all four motors individually before any strafe math; do not let them skip it. 135–160: Run Fast Lane. Everyone drives at least one lap. Count laps and penalties, enter them in the Skill Builders scoring sheet.

If you fall behind. The two checkpoints that decide everything are **BA-2 finished Sept 18** and **BA-3 finished Sept 22**. If either slips, move people off the media and merch cards and onto the build for the next meeting — those cards keep, the deadline does not. If Sept 25 arrives and the robot does not drive, still run Fast Lane: tape the track, walk the chassis around it by hand, and let the drivers practice on the dev board instead. **Forward and turning alone are enough for Fast Lane** — if strafing is fighting you on the 25th, ship the tank-style version and fix strafing after break. Then finish the build first thing after break.

Before you walk out on Sept 25. Two weeks is long enough to forget everything. Leave the robot in a known state — nothing half-disassembled, no loose wiring — get every crew's notebook entry written (**EP-5**), and photograph the shop matching the BA-1 reference shelf photo.

Hardware & the second robot

Coach page — what can run at the same time

The hardware column — how to run two tracks at once

Every card is tagged with the hardware it needs. This is the constraint that decides what can run in parallel: **32 of the 53 cards need no hardware at all**, so there is always something for a crew to pick up.

TAG	CARDS	WHAT IT MEANS FOR PLANNING
No hardware	32	Runs anywhere, any number of crews at once. Planning, all seven CAD cards , merch, shirts, marketing, plus PR-9 (simulator) and BA-1/BA-6. <i>CD-6 and CD-7 need printer access — a shared resource, but not one that competes for the robot.</i>
Programming board	4	Needs the 3D-printed board. BA-9 builds it; PR-10, PR-11 and PR-12 are the sensor cards only the board can teach — touch, color and potentiometer are not on the StarterBot base at all.
Board or robot	3	Either one works. Run these on the board while the build crew still has the robot apart.
Robot / Robot + field	14	The scarce resource — one crew at a time, unless the class robot is both present and has its own hub. “Robot + field” also needs EP-6 done first.

Hub accounting — settle this before Sept 18. The **Mecanum Drivetrain Kit v2** does not include a **Control Hub**, a battery, or any wiring. The team has two hubs: **one is on the programming board**, which leaves **one for the student robot**. That accounts for both, and it is exactly what the two-track plan needs — neither ever gets unbolted to run a meeting. But confirm the second hub is genuinely free and not committed to the Herobotix class robot. **No hub on the student robot means no Fast Lane on Sept 25**, so if a third is needed, it has to be ordered with enough lead time to arrive and be tested. A third hub would also give the class robot back as a second driving station.

When the Herobotix class robot is available

It is shared, so plan every meeting around one robot and treat the second as a bonus. It runs the **same REV Control Hub**, so OpModes port over — but a configuration saved on one hub does not exist on the other, so it needs its own config built (PR-2) before anything runs. **First check it still has a hub** (see hub accounting above). On the meetings it shows up ready, these are the cards worth reaching for:

CARD	WHY A SECOND ROBOT CHANGES IT
PR-5 Driver Tuning	Two drivers practice at once instead of taking turns. Doubles the reps in the same 100 minutes, and the penalty counts become comparable between drivers.
EP-6 Field — Fast Lane	The Skill Builders deck suggests a wider track for multiple robots and a Grand Prix format. That needs two robots to exist.
PR-6 / PR-7 Autonomous	One robot runs while the other crew edits code, instead of both crews queuing for the same robot between test runs.
PR-13 Bench to Robot	Port the same bench behavior onto both robots. Whatever breaks on the second one is the part the students did not actually understand.
MF-6 Demo Event	Two robots make a far better demo, and lets more students drive in front of an audience.

Team Identity: Name, Look & Motto

No hardware

Start here

3-5 people

100 min

WHY THIS MATTERS

Everything downstream needs a name and a look — shirts, buttons, sponsor letters, the website. Until this is done, the merch and marketing tracks are blocked. Do it first, do it once.

YOU WILL FINISH WITH

One team name, one backup, a written identity brief, and three rough logo directions on paper.

WHAT YOU NEED

- Whiteboard or big paper
- Markers
- Sticky notes
- A phone or laptop to search names

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

MF-2, MF-4, MF-5, MG-1, MG-2, MG-4, MG-6, TS-1

IF YOU FINISH EARLY

- Write a 15-second team introduction that any member could say to a judge or a sponsor without notes.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Get a big writing surface and hand everyone a marker.
- ☐ Look up the other Herobotix team names so you do not pick one that is taken.
- ☐ Write the season name (BIOBUZZ) and the scoring element (Pollen) on the board — both are good idea fuel.

15-45 min

- ☐ SILENT round: everyone writes 5 name ideas on sticky notes. No talking, no judging. 5 minutes.
- ☐ Post every note on the board. Read them all out loud.
- ☐ Group similar ideas together. Still no criticism — that comes next.

45-65 min

- ☐ Score each idea out loud: Is it easy to say? Does it fit on a shirt? Can you draw it? Is it already taken (search it)?
- ☐ Cut the board down to 3 finalists.

65-85 min

- ☐ Vote. Everyone gets one vote. Pick a winner and a backup.
- ☐ Write both on the board and photograph it.

85-100 min

- ☐ Write the identity brief — 3 sentences: who we are, what we are about, what our look is.
- ☐ Name 2-3 team colors and one shape or mascot idea.
- ☐ Sketch 3 rough logo directions. Rough is fine. Tape them up.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Name and backup written down and photographed
- ☐ 3-sentence identity brief in the engineering notebook
- ☐ 3 logo sketches taped to the wall

HAND OFF TO

Hand the identity brief and logo sketches to whoever takes TS-1 (shirt design) and MG-2 (give-away art).

Who worked on this: _____ Date: _____

Read the Game, Build the Wall Poster

No hardware

Start here

2-4 people

100 min

WHY THIS MATTERS

Every decision this season traces back to the Competition Manual. If nobody on the team has actually read the scoring section, you will build the wrong robot and find out in January.

YOU WILL FINISH WITH

A one-page game summary poster on the wall that anyone can point at mid-argument.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-10 min

- ☐ Open the Competition Manual and find the scoring section.
- ☐ Split the scoring rules among the crew — one person per section.

10-40 min

- ☐ Read your section. Write down every way to score, what it is worth, and when it is allowed (autonomous / driver-controlled / endgame).
- ☐ Write down anything you do not understand — you will ask the coach.

40-70 min

- ☐ Build the poster: a sketch of the field, a scoring table, and a match timeline bar showing each phase and how many seconds it lasts.

70-90 min

- ☐ Add a GOTCHAS box: the top 5 penalties and what each one costs.

90-100 min

- ☐ Present the poster to the rest of the team in 3 minutes. Hang it where the robot gets built.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Competition Manual (printed or on a laptop)
- Poster paper
- Markers
- Ruler

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

CD-4, EP-3

IF YOU FINISH EARLY

- Add a points-per-second column. Which scoring method is actually worth chasing when the clock is running?

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Poster is on the wall where the build happens
- ☐ Scoring table matches the Competition Manual exactly
- ☐ Anyone on the team can name the three match phases and their lengths

HAND OFF TO

The build and programming crews will use this poster constantly. Tell them it exists.

Who worked on this: _____ Date: _____

Robot Requirements List

WHY THIS MATTERS

'Build a robot' is not a plan. A requirements list turns the rulebook into specific things you can build and then actually check off.

YOU WILL FINISH WITH

A numbered requirements list, each labeled MUST / SHOULD / NICE, each with a way to test it.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pull up the game poster and the robot/inspection rules in the Competition Manual.

15-45 min

- ☐ Write requirements as testable sentences. Good: 'Robot fits inside the legal starting size.' 'Robot picks up one Pollen off the floor.' Bad: 'Robot is fast.'
- ☐ Aim for 15-25 requirements. Number them.

45-65 min

- ☐ Label each one MUST, SHOULD, or NICE. Argue about it — the argument is the valuable part.
- ☐ MUST means the robot is illegal or useless without it.

65-85 min

- ☐ Add a HOW WE TEST IT column to every requirement. If you cannot describe the test, it is not a requirement yet — rewrite it.

85-100 min

- ☐ Post it next to the game poster. Walk the build crew through it.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ List is numbered and every item is labeled MUST / SHOULD / NICE
- ☐ Every MUST has a written test
- ☐ Posted next to the game poster

WHAT YOU NEED

- The EP-2 game poster
- Competition Manual (inspection + robot rules section)
- Laptop or large paper

BEFORE YOU START

EP-2

THIS UNLOCKS

EP-4

IF YOU FINISH EARLY

- Add a STATUS column (Not started / In progress / Passes). This list is now your build tracker for the whole season.

HAND OFF TO

This is the scorecard for EP-4 concept selection and for every build task.

Who worked on this: _____ Date: _____

Design Sketches & Pick a Concept

No hardware

Some experience

3-4 people

100 min

WHY THIS MATTERS

The first idea is rarely the best one, and the loudest person's idea is not automatically right. Sketching three and scoring them takes an hour and can save weeks of rebuilding.

YOU WILL FINISH WITH

Three sketched concepts for one mechanism, scored against your requirements, one chosen with a written reason.

WHAT YOU NEED

- Paper and pencils (one sheet each)
- The EP-3 requirements list
- Whiteboard

BEFORE YOU START

EP-3

THIS UNLOCKS

BA-5, BA-7, CD-3

IF YOU FINISH EARLY

- Sketch the backup plan and write down what would have to go wrong before you switch to it.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-10 min

- ☐ Pick ONE mechanism to design today: the intake, the scoring/outtake, or how Pollen gets stored on the robot. One. Not all three.

10-40 min

- ☐ Everyone sketches independently. No looking at each other's paper. Label the parts. Draw at least 2 views (side and top).

40-55 min

- ☐ Pin up all sketches. Each person explains theirs in 90 seconds. Listeners ask questions, not criticisms.

55-80 min

- ☐ Build a decision matrix on the whiteboard. Rows = concepts. Columns = 'Can we build it with parts we own?', 'How many meetings to build?', 'How likely to break?', 'Points it earns'.
- ☐ Score every box 1 to 5. Total the columns.

80-95 min

- ☐ Pick the winner. Write one paragraph explaining WHY — including what you gave up by not picking the others.

95-100 min

- ☐ Photograph every sketch and the matrix into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ 3 or more labeled sketches with 2 views each
- ☐ Decision matrix filled in and totaled
- ☐ A written paragraph explaining the choice and the trade-off

HAND OFF TO

The chosen concept goes to BA-7 (prototype) and into the engineering notebook.

Who worked on this: _____ Date: _____

Set Up the Engineering Notebook

No hardware

Start here

2-3 people

100 min

WHY THIS MATTERS

Judges read the notebook. More importantly, it is how you remember in February why you made a choice back in October. A team without a notebook resolves the same problem three times.

YOU WILL FINISH WITH

A working notebook with sections, a reusable daily template, and today's entry already written.

WHAT YOU NEED

- Binder and dividers, or a shared doc (ask the coach which the team is using)
- Printer access
- Photos from past meetings

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add at least one photo to every back-filled entry.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Ask the coach what format the team is using: physical binder, shared doc, or both. Set it up.

15-35 min

- ☐ Build the sections: Team & Roles / Meeting Log / Design (one per mechanism) / Programming / Outreach / Business & Budget.

35-55 min

- ☐ Make the one-page daily entry template: Date | Who was here | What we set out to do | What we actually did | What we learned | What is next.
- ☐ Print 20 copies, or duplicate the page in the shared doc 20 times.

55-80 min

- ☐ Back-fill an entry for every meeting so far, using photos and memory. Do not skip this — it only gets harder every week.

80-95 min

- ☐ Write today's entry properly, as the example everyone else will copy.

95-100 min

- ☐ Write the name of next meeting's notebook owner on the whiteboard.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ All sections exist
- ☐ Daily template exists and is stocked
- ☐ Every past meeting has an entry
- ☐ Today's entry is written
- ☐ Next owner named

HAND OFF TO

Notebook duty rotates. Whoever is named owns capturing the next meeting.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

You cannot practice driving or test an autonomous routine without a marked field. This is also the setup for the Fast Lane and Wayfinder Skill Builders challenges.

YOU WILL FINISH WITH

A taped practice area with correct, measured dimensions, ready for driving challenges.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ ASK THE COACH before taping anything — confirm the floor is OK and only painter's or gaffer's tape is used.
- ☐ Measure the room. Find the largest clear rectangle you can use.

15-30 min

- ☐ Mark the four corners. Measure both diagonals — if they are equal, your rectangle is square. Adjust until they match.

30-60 min

- ☐ Lay the outer boundary tape. Then lay a Fast Lane track inside it, at least 2 ft (0.61 m) wide everywhere.
- ☐ Track shape is your choice — see the Fast Lane slides for oval and figure-8 options.

60-80 min

- ☐ Mark a Start/Finish line, an Alliance Area off the field where drivers stand, and at least 4 checkpoints for Wayfinder.

80-95 min

- ☐ Drive a robot around it (or walk the path holding a 18 in box). Fix any corner that is too tight.

95-100 min

- ☐ Photograph the layout and write the measurements onto the photo.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Diagonals measured equal
- ☐ Track is at least 2 ft wide the whole way around
- ☐ Start/Finish, Alliance Area, and 4+ checkpoints marked
- ☐ Measured photo in the notebook

WHAT YOU NEED

- Painter's or gaffer's tape in a high-contrast color
- Tape measure (25 ft)
- The Fast Lane and Wayfinder challenge slides

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

PR-5, PR-6

IF YOU FINISH EARLY

- Build a simple removable border out of PVC or 1x4 lumber so the robot cannot run off the field.

HAND OFF TO

Tell the programming crew the field is up — PR-5, PR-6 and PR-7 all need it.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Nobody touches a tool until they are checked off. It is also the fastest way for a brand-new member to go from watching to being useful.

YOU WILL FINISH WITH

Every member signed off on the safety rules and on each hand tool they are cleared to use.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Coach walks the whole group through the space: exits, first aid kit, fire extinguisher, where safety glasses live, where batteries live.

15-35 min

- ☐ As a group, write the team's OWN safety rules in your own words — 8 to 12 of them. Post them on the wall.

35-65 min

- ☐ HAND TOOL stations — hex drivers, wrenches, cutters, files. One person demonstrates, then EVERY member does one supervised repetition.
- ☐ These are the tools the build cards actually call for, so everyone gets cleared on them today.

65-85 min

- ☐ POWER TOOLS are a different thing and today you do not run them. Walk to the chop saw, scroll saw, belt sander and drill press as a group.
- ☐ The coach names each one, shows where its switch and its stop are, and explains that it stays off-limits until you are individually signed off by a mentor.
- ☐ Read the Power Tool Check-Off poster out loud together — especially the line that a signature never means you may use it alone.

85-100 min

- ☐ Fill in the hand-tool grid and post both sheets where the tools live.
- ☐ Write down who still needs clearing. Power tool sign-offs happen later, two or three students at a time with a mentor — never a whole group in one session.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Team-written safety rules posted
- ☐ Hand-tool grid filled in for everyone present and posted
- ☐ Power Tool Check-Off poster up, read aloud, and understood — nobody ran a power tool today
- ☐ Safety glasses counted and located
- ☐ A written list of who still needs clearing

WHAT YOU NEED

- Safety glasses for everyone
- The team's hand tools
- The printed hand-tool check-off grid
- The printed Power Tool Check-Off poster
- Coach or mentor to lead

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

BA-2, BA-4, BA-9

IF YOU FINISH EARLY

- Make a laminated one-page safety card to live in the pit toolkit at competitions.

HAND OFF TO

The build tasks assume the hand-tool clearing is done. Power tool sign-offs are a separate, ongoing thing a mentor does in small groups — keep both grids current as new members join.

Who worked on this: _____ Date: _____

Onshape Setup & Your First Part

No hardware

Start here

2-4 people

100 min

WHY THIS MATTERS

Onshape is free for FIRST teams, runs in a browser on any computer including a Chromebook, and it is what professional engineers actually use. Everything the team designs from here lives in it.

YOU WILL FINISH WITH

Everyone has an account in the team workspace and has modeled a real part with exact dimensions.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Everyone creates a free Onshape account through the FIRST education link. The coach approves the signups — students should not be creating accounts unsupervised.
- ☐ Join the team's shared Onshape workspace so your work is visible to everyone.

20-45 min

- ☐ Interface tour together: the Part Studio, a sketch, the extrude tool, and the feature tree down the left side.
- ☐ Follow one intro tutorial as a group before anyone works alone.

45-80 min

- ☐ Everyone models the SAME part so you can compare: a flat plate 100mm x 60mm x 6mm with two 5mm holes 70mm apart.
- ☐ Sketch it on a plane, dimension it properly (do not eyeball it), extrude it, then cut the holes.
- ☐ The loop you are learning is sketch, dimension, extrude. Everything else is a variation of it.

80-95 min

- ☐ Now change the plate's length from 100mm to 130mm and watch the whole part update. That is what makes CAD different from drawing a picture.

95-100 min

- ☐ Share your part with the coach. Write down the three tools you used most.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Account created and joined to the team workspace
- ☐ Plate modeled to the exact dimensions given, not eyeballed
- ☐ A dimension changed and the model updated correctly
- ☐ Part shared with the coach

WHAT YOU NEED

- One computer per person
- The Onshape for FIRST link on the resource shelf
- Coach present to approve account signups
- A ruler for reference

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

CD-2

IF YOU FINISH EARLY

- Model the same plate again from scratch without looking at notes. In CAD, speed is entirely repetition.

HAND OFF TO

CD-2 is the next step. Tell the team the workspace exists and who is in it.

Who worked on this: _____ Date: _____

Measure & Model a Real Robot Part

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

CAD is only useful if it matches reality. Modeling something you can hold, then checking your model against it with calipers, is how you learn to trust your own drawings — and how you catch that you have been guessing.

YOU WILL FINISH WITH

An accurate Onshape model of a real part from the kit, verified against the physical part.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pick a real part with a few holes in it. Do not pick the most complicated thing on the shelf for your first one.
- ☐ Get calipers and learn to read them — ask a mentor if you have not used them.

15-40 min

- ☐ Measure it properly and write EVERY dimension on paper before you model anything.
- ☐ Measure each dimension twice. If the two readings disagree, measure a third time and find out why.

40-80 min

- ☐ Model it in Onshape from your written measurements — not from looking at the part on the table.

80-95 min

- ☐ Check your work: compare the model's dimensions against the physical part with calipers. Anything more than about 0.5mm off, find out why before you move on.

95-100 min

- ☐ Save it into the team's parts library with a name someone else would recognize.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ A written measurement sheet made BEFORE modeling started
- ☐ Model built from the sheet
- ☐ Every dimension verified against the real part within about 0.5mm
- ☐ Saved to the team library with a clear name

WHAT YOU NEED

- Onshape account (CD-1)
- Calipers (ask the coach)
- A real part: a C-channel offcut, a bracket, a spacer, a wheel hub
- Paper for the measurement sheet

BEFORE YOU START

CD-1

THIS UNLOCKS

CD-3, CD-4, CD-5, CD-6

IF YOU FINISH EARLY

- Model a second part and mate the two together in an Assembly. That is the skill CD-5 is built on.

HAND OFF TO

The team parts library is what makes CD-3, CD-4 and CD-5 fast. Keep adding to it.

Who worked on this: _____ Date: _____

Design a Pollen Intake in CAD

WHY THIS MATTERS

The Mecanum Drivetrain Kit is a drive base and nothing else, so the intake is entirely your team's design. Working it out in CAD first means you find the problems before anyone cuts metal or burns three hours of filament.

YOU WILL FINISH WITH

A CAD model of an intake concept, sized against a real Pollen and the real mounting space.

WHAT YOU NEED

- Onshape (CD-2)
- The concept chosen in EP-4
- Calipers
- A real Pollen to measure
- Access to the robot to measure the mounting space

BEFORE YOU START

CD-2, EP-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Model a second, genuinely different concept and put both in front of the team. Two options make a better decision than one.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Get real numbers before you draw anything: measure a Pollen with calipers, and measure the space on the robot where this has to mount. Write both down.
- ☐ Review the concept the team chose in EP-4 — you are modeling that, not starting a new argument.

20-50 min

- ☐ Model the working part first — the thing that actually touches the Pollen. Keep it simple; simple things get built.

50-75 min

- ☐ Model the mounting. How does it attach to the C-channel? Measure the channel's real hole spacing and match it exactly, or it will not bolt on.

75-95 min

- ☐ Sanity-check in CAD: does a Pollen-sized object fit through the opening? Is anything thinner than 3mm, which will snap? Does it stick out past the robot's legal starting size?

95-100 min

- ☐ Share it with the build crew and write down honestly what you are unsure about.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Pollen and mounting space both measured and written down
- ☐ Intake modeled with mounting holes that match the real channel spacing
- ☐ Clearance checked against a Pollen-sized object in CAD
- ☐ Shared with the build crew with open questions noted

HAND OFF TO

BA-5 builds it. CD-6 can print parts of it.

Who worked on this: _____ Date: _____

Design a Pollen Outtake in CAD

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

Getting Pollen in is one problem; putting it where you want it is a different one. An outtake has to release on command, aim, and not jam — and that geometry is far easier to try in CAD than in aluminum.

YOU WILL FINISH WITH

A CAD model of an outtake concept with the Pollen's path through it checked.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Look at the EP-2 game poster: where does Pollen actually have to go, and how high? Write those numbers down — they are the requirements.

20-45 min

- ☐ Sketch the Pollen's path through the mechanism ON PAPER first: where it enters, where it waits, where it exits, and what pushes it out.

45-80 min

- ☐ Model it. Model a simple Pollen shape too, so you can place it inside your mechanism and actually see whether the path works.

80-95 min

- ☐ Check it: does anything block the exit? Is the release repeatable, or does it depend on the Pollen sitting perfectly? Mark where it would jam.

95-100 min

- ☐ Share with the build crew and note what you are unsure about.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Scoring target and height recorded from the game poster
- ☐ Pollen path sketched on paper before modeling
- ☐ Mechanism modeled with a Pollen positioned inside it
- ☐ Likely jam points identified and noted

WHAT YOU NEED

- Onshape (CD-2)
- The EP-2 game poster for scoring targets and heights
- Calipers and a real Pollen
- Paper for the path sketch

BEFORE YOU START

CD-2, EP-2

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Model it in two positions, loaded and released, so the team can see the motion instead of imagining it.

HAND OFF TO

Hand it to BA-7, which prototypes in cardboard. CAD and cardboard together beat either one alone.

Who worked on this: _____ Date: _____

Build the Robot in CAD: Assembly & Fit Check

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

If the robot exists in CAD, you can find out whether a new mechanism fits before anyone builds it. REV publishes the Mecanum Drivetrain CAD, so you do not have to model the chassis yourself.

YOU WILL FINISH WITH

The drivetrain in an Onshape assembly with a team-designed part mated to it, and a real fit check done.

WHAT YOU NEED

- Onshape (CD-2)
- The Mecanum Drivetrain V2 build guide page, which links its Onshape and STEP CAD
- A part your team designed (CD-3, CD-4 or CD-7)
- Tape measure to check against the real robot

BEFORE YOU START

CD-2

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Measure the robot's total width and length in CAD, then measure the real robot with a tape measure. If they disagree, one of them is wrong and it is worth knowing which.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Find the Mecanum Drivetrain V2 CAD linked from REV's build guide page. Copy it into the team workspace.

20-45 min

- ☐ Create an Assembly and bring the drivetrain in. FIX it in place first, or everything you add will float around and drive you crazy.

45-75 min

- ☐ Bring in a part your team designed and mate it to the chassis exactly where it would really mount.

75-95 min

- ☐ Fit check. Does it collide with a wheel through the wheel's full rotation? Does it stick past the legal starting size? Use Onshape's measure tool on the overall dimensions.

95-100 min

- ☐ Screenshot the assembly into the notebook and tell the build crew what you found — especially if it does not fit.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ REV drivetrain CAD imported into the team workspace
- ☐ Assembly created with the chassis fixed in place
- ☐ A team-designed part mated where it would really mount
- ☐ Overall size measured in CAD and findings reported to the build crew

HAND OFF TO

From now on, any new mechanism gets a fit check here before the build crew cuts anything.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

A model on a screen is a drawing until it exists. Slicing and printing teaches you what CAD cannot: overhangs sag, thin walls snap, and holes always print undersized.

YOU WILL FINISH WITH

One part designed in Onshape, printed, and measured against the model to find your printer's real offset.

WHAT YOU NEED

- A model from the team library (CD-2) or your own
- 3D printer access — confirm with the coach
- Filament
- Slicer software
- Calipers

BEFORE YOU START

CD-2

THIS UNLOCKS

CD-7

IF YOU FINISH EARLY

- Reprint the same part with the hole diameter corrected by your offset, and confirm the bolt actually fits this time.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Confirm printer access and filament with the coach. Pick a SMALL part — under two hours of print time. Ambition here costs days.

15-35 min

- ☐ Export from Onshape as an STL and open it in the slicer.

35-55 min

- ☐ Set it up: layer height, infill, supports if the shape needs them, and which face goes down on the plate. Read the estimated time.
- ☐ Write every setting in the notebook. A print you cannot reproduce taught you nothing.

55-75 min

- ☐ Start the print. If it runs past the meeting, that is fine — the coach can pull it and you finish this card next meeting.

75-95 min

- ☐ Measure the printed part against the CAD model with calipers. The holes will almost certainly be undersized.
- ☐ Write down the difference. That number is your printer's real-world offset.

95-100 min

- ☐ Record the settings, the time, and the measured offset in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ STL exported and sliced with every setting written down
- ☐ Part printed
- ☐ Printed dimensions compared to the model with calipers
- ☐ The printer's offset recorded

HAND OFF TO

Write the offset number big on the whiteboard. Every part the team prints from now on should use it.

Who worked on this: _____ Date: _____

Design a Printed Part That Solves a Real Problem

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

The best use of a 3D printer on an FTC team is not decorations. It is the bracket nobody sells. Find something the robot actually needs and make it exist.

YOU WILL FINISH WITH

A printed part installed on the robot that fixes a problem the team really has.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Walk to the robot with the build crew and ask out loud: what is being held on with tape, zip ties, or hope? What bracket would we buy if it existed?
- ☐ Write down three candidates. Do not pick yet.

20-35 min

- ☐ Pick the one that is most annoying and least complicated. Measure everything it has to attach to.

35-70 min

- ☐ Model it, designing for printing this time: avoid steep overhangs, keep walls at least 3mm, put material where the load actually goes, and apply your printer offset to every hole.

70-85 min

- ☐ Slice it, check the time, and print it or queue it with the coach.

85-100 min

- ☐ Write the case for it in the notebook: what problem it solves, what it replaces, how long it took to print.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- CD-6 done, including the printer offset
- Calipers
- Access to the robot with the build crew
- Printer access

BEFORE YOU START

CD-6

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Photograph the before and after once it is installed. This exact story — problem, design, part, result — is what judges ask for.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Three real problems listed from looking at the actual robot
- ☐ One chosen and fully measured
- ☐ Part modeled with printability and the printer offset accounted for
- ☐ Printed or queued, with a written case for why it exists

HAND OFF TO

This is a portfolio piece. Make sure EP-5 captures it properly in the notebook.

Who worked on this: _____ Date: _____

Get Connected: Control Hub & Driver Station

Board or robot

Start here

2-3 people

100 min

WHY THIS MATTERS

Nothing else in programming happens until the Driver Station talks to the Control Hub. Do this once, carefully, and write down exactly what you did so nobody has to rediscover it.

YOU WILL FINISH WITH

Driver Station paired to the Control Hub, robot showing as connected, and the programming editor open in a browser.

WHAT YOU NEED

- REV Control Hub
- Driver Hub or Driver Station phone
- Charged 12V battery
- USB cable
- Laptop
- Mecanum Drivetrain V2 build guide + FTC Docs

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

PR-2

IF YOU FINISH EARLY

- Set up a second Driver Station and practice swapping between them, so a dead phone never ends your day.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Gather everything on the list. Confirm the battery is actually charged — most first-day problems are a dead battery.
- ☐ Open the FTC Docs programming pages. If you are connecting the robot rather than the dev board, have the Mecanum Drivetrain V2 build guide open too.

15-35 min

- ☐ Power the Control Hub from the battery. Watch the status lights and look up what the colors mean.
- ☐ Find the Control Hub's WiFi network name.

35-55 min

- ☐ On the Driver Station, connect to the Control Hub's network.
- ☐ CHANGE THE DEFAULT WIFI PASSWORD and write the new one in the engineering notebook. Tell the coach what it is.

55-75 min

- ☐ On the laptop, join the same network and open the Blocks / OnBot Java editor in a browser. The address is in FTC Docs.

75-90 min

- ☐ Check the Control Hub firmware version and the SDK version. Write both in the notebook. Ask the coach whether an update is needed.

90-100 min

- ☐ Write a half-page 'How to connect' guide with the exact steps you just did. Tape it inside the laptop case.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Driver Station shows the robot connected
- ☐ Browser opens the programming editor
- ☐ New WiFi password recorded
- ☐ Firmware + SDK versions recorded
- ☐ Connect guide written

HAND OFF TO

PR-2 cannot start until this works. Tell the next crew where the connect guide is.

Who worked on this: _____ Date: _____

Build the Robot Configuration File

Board or robot

Start here

2-3 people

100 min

WHY THIS MATTERS

The code cannot find a motor that is not in the configuration. Roughly half of all 'my code does not work' problems are a typo in this file.

YOU WILL FINISH WITH

A saved, activated configuration that matches the physical robot, and a port map taped to the robot.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ With the build crew, list every motor and servo: what it does and which hub port it is plugged into. Write it on the whiteboard.

15-35 min

- ☐ Agree on names and be boring about it: lowercase with underscores, like `left_front_drive`, `right_front_drive`, `intake_motor`.
- ☐ Post the naming list. Everyone uses these exact names from now on.

35-70 min

- ☐ On the Driver Station: Configure Robot, create a new configuration, add each device to its correct port with its exact name.
- ☐ Save it with the team name in the filename.

70-85 min

- ☐ Activate the configuration. Spin each motor for one second and confirm the RIGHT motor moves. Fix any mismatch now.

85-100 min

- ☐ Print or hand-write a port map and tape it to the robot. Photograph it into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every named motor spins the correct physical motor
- ☐ Configuration saved AND activated
- ☐ Port map taped to the robot
- ☐ Naming list posted

WHAT YOU NEED

- Connected robot (PR-1)
- The build crew, or their port notes
- Whiteboard
- Label tape

BEFORE YOU START

PR-1

THIS UNLOCKS

PR-10, PR-3, PR-8

IF YOU FINISH EARLY

- Note which motors will need to be reversed in code for the robot to drive straight — PR-3 will need that.

HAND OFF TO

Give the naming list to everyone who writes code. Changing a name later breaks every OpMode.

Who worked on this: _____ Date: _____

PR-3

First TeleOp: Make It Drive (Mecanum)

WHY THIS MATTERS

This is the moment robotics gets real. Mecanum means the robot can drive, turn, AND slide sideways without turning — but only if all four corners are commanded correctly. Everything else this season builds on this program.

YOU WILL FINISH WITH

A driver-controlled OpMode that drives, turns, and strafes, plus a saved known-good copy.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-10 min

- ☐ Open Blocks and create a new TeleOp OpMode named TeleOp_Mecanum_v1.

10-30 min

- ☐ PROVE THE MOTORS FIRST. Before any drive math, write a program that runs ONE motor at a time on four different buttons.
- ☐ Prop the robot up, press each button, and confirm the right corner spins the right way. Fix any direction here.
- ☐ Skipping this step is the number one reason mecanum 'does not work'.

30-60 min

- ☐ Add forward, backward and turning only — no strafing yet. Left stick forward/back drives, right stick left/right turns. Test it on the floor.

60-85 min

- ☐ Now add strafing. With drive = left stick Y, strafe = left stick X and turn = right stick X, the four powers are: front-left = drive + strafe + turn; front-right = drive - strafe - turn; back-left = drive - strafe + turn; back-right = drive + strafe - turn.
- ☐ Then find the largest absolute value of the four; if it is over 1, divide all four by it so nothing gets clipped.
- ☐ Test strafing left and right. If the robot spins instead of sliding, either a sign is wrong or a wheel is in the wrong corner — go back to BA-3 and re-check the X.

85-95 min

- ☐ Save. Then save a second copy named TeleOp_Mecanum_v1_WORKING so you always have something that runs.

95-100 min

- ☐ Write in the notebook which motors are reversed and how the strafe test went.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Robot drives forward and backward and turns both directions
- ☐ Robot strafes left and right without rotating
- ☐ A known-good copy is saved under a separate name

WHAT YOU NEED

- Configured robot with four drive motors (PR-2)
- Gamepad
- Blocks in the browser
- Something to prop the robot on with the wheels off the ground
- Clear floor space

BEFORE YOU START

PR-2

THIS UNLOCKS

MF-6, PR-13, PR-14, PR-4, PR-5, PR-6

IF YOU FINISH EARLY

- Drive a square around the taped field without ever turning — forward, strafe, forward, strafe. If you can do that, your mixing is right.

HAND OFF TO

Tell the whole team — the robot drives now. This unlocks driver practice, Fast Lane, and demo events.

Who worked on this: _____ Date: _____

Telemetry: See What the Robot Is Thinking

Board or robot

Some experience

1-2 people

100 min

WHY THIS MATTERS

Telemetry prints live values to the Driver Station screen. Without it you are guessing at why something failed. With it, you can see the problem.

YOU WILL FINISH WITH

The Driver Station shows live, correct values for sticks, motor power, encoders, and time.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the telemetry section in the FTC Docs programming resources.

15-45 min

- ☐ Add telemetry lines for: each gamepad stick value, each motor's commanded power, and elapsed match time.

45-70 min

- ☐ Run it while someone drives and watch the screen. Do the numbers match what you expect?
- ☐ Check that the sticks read 0 when centered — if not, you have found a real problem worth fixing (see PR-5).

70-90 min

- ☐ Add encoder position readouts for the drive motors. You will need these for PR-7.

90-100 min

- ☐ Photograph the Driver Station screen into the notebook with a note about what each line means.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Working TeleOp (PR-3)
- FTC Docs telemetry section
- Driver Station

BEFORE YOU START

PR-3

THIS UNLOCKS

PR-11, PR-12, PR-14, PR-7

IF YOU FINISH EARLY

- Add a line that reads battery voltage and shouts a warning when it drops below a threshold you pick.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Sticks, power, encoders, and time all display and change correctly
- ☐ Centered sticks read at or near 0
- ☐ Screen photographed with notes

HAND OFF TO

Every later programming task should keep telemetry in. Tell the next crew not to delete it.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

New drivers over-steer — everyone does. A small amount of code makes a huge difference in Fast Lane scores and in real matches. This is measurable improvement in one meeting.

YOU WILL FINISH WITH

A drive OpMode with a precision mode and smoother stick response, with before/after penalty counts to prove it worked.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Drive the Fast Lane track once. Count every line crossed and wall hit. WRITE THE NUMBER DOWN — this is your baseline.

15-45 min

- ☐ Add a precision mode: while a trigger or bumper is held, scale motor power down to about 30-40 percent.

45-70 min

- ☐ Add a deadband so tiny stick movements near center are ignored.
- ☐ Try squaring the stick input (keep the sign) so small movements give finer control and full push still gives full speed.

70-90 min

- ☐ Re-run the track with the same driver. Count penalties again. Compare to the baseline.

90-100 min

- ☐ Record both numbers and exactly what you changed in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Working TeleOp (PR-3)
- Taped practice track (EP-6)
- Stopwatch
- Someone to count penalties

BEFORE YOU START

PR-3, EP-6

THIS UNLOCKS

IF YOU FINISH EARLY

- Add a reverse-view toggle that flips the controls for when the driver is driving the robot toward themselves.
- The Fast Lane deck has a mecanum-only extension: drive the entire track with the robot facing one direction the whole way, using strafing instead of turning. It is much harder and it is superb driver practice.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Precision mode works on a held button
- ☐ Deadband stops phantom drift
- ☐ Before and after penalty counts recorded, with the change written down

HAND OFF TO

Hand the tuned OpMode to the drivers. Note which settings they liked.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Autonomous is just a list of instructions with timing. Start here before anything clever. This is the Wayfinder challenge in its simplest form.

YOU WILL FINISH WITH

An autonomous OpMode that drives a repeatable path with no driver input, and a measurement of how repeatable it actually is.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Draw the path on paper FIRST, as numbered steps: forward 2.0 s, stop, turn right 0.8 s, stop, forward 1.5 s, stop.

15-40 min

- ☐ Create a new Autonomous OpMode. Build **ONLY** the first move: drive forward, then stop. Test it three times.

40-70 min

- ☐ Add one step at a time. Test after EVERY single step. Change one number at a time — never two.

70-90 min

- ☐ Run the complete routine 5 times from the exact same starting spot. Measure how far off the ending position is each time and write down all 5.

90-100 min

- ☐ Record the final step list and the spread of the 5 runs in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Working drive code (PR-3)
- Taped field (EP-6)
- Tape measure
- Paper for the step plan

BEFORE YOU START

PR-3, EP-6

THIS UNLOCKS

PR-7

IF YOU FINISH EARLY

- Run it on a fresh battery and again on a low battery. Measure the difference. That gap is exactly why encoders exist — see PR-7.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Robot runs the full path unassisted 3 times in a row
- ☐ 5 runs measured and recorded
- ☐ Step list written in the notebook

HAND OFF TO

The measured spread tells the team whether time-based autonomous is good enough or whether PR-7 is needed.

Who worked on this: _____ Date: _____

Encoder Autonomous: Drive an Exact Distance

Robot + field

Challenge

2-3 people

100 min

WHY THIS MATTERS

Time-based autonomous drifts as the battery drains, so it works in practice and fails in a match. Encoders count actual wheel rotations, so the distance stays honest.

YOU WILL FINISH WITH

A routine that drives a commanded number of inches, landing within about an inch, five times running.

WHAT YOU NEED

- Working time-based auto (PR-6)
- Encoder telemetry (PR-4)
- Tape measure
- Motor spec sheet
- Calculator
- gm0 + FTC Docs

BEFORE YOU START

PR-6, PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add an encoder-based turn, or use the Control Hub's built-in IMU to turn an exact number of degrees.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the encoder section in FTC Docs and gm0.
- ☐ Find your motor's counts per revolution (check the REV spec sheet for your exact motor) and measure your wheel diameter.

20-40 min

- ☐ Calculate counts per inch: (counts per motor revolution multiplied by gear ratio) divided by (wheel circumference in inches).
- ☐ Show the whole calculation on the whiteboard so someone else can check it.

40-75 min

- ☐ Write a drive-to-distance routine using RUN_TO_POSITION, or by reading encoder counts and stopping at the target.

75-95 min

- ☐ Command exactly 24 inches. Measure the real distance with a tape measure. Repeat 5 times, recording every result.
- ☐ Adjust your counts-per-inch constant based on what you measured, then re-test.

95-100 min

- ☐ Write the final counts-per-inch number large on the whiteboard and in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Commanded 24 in lands within about 1 in, five times in a row
- ☐ The counts-per-inch math is shown and checked
- ☐ Final constant recorded

HAND OFF TO

This constant is a team asset. Anyone writing autonomous should be able to find it in 10 seconds.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

A driver has to run the intake without looking down at the controller. Getting the button mapping and the power level right is what makes the difference between grabbing Pollen and throwing it.

YOU WILL FINISH WITH

Intake runs in and out on gamepad buttons at a tested power level, with a control map taped to the Driver Station.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm the intake motor is in the configuration and spins the correct direction. Fix it in the config, not in code, if it is backwards.

15-45 min

- ☐ Map intake-in and intake-out to two buttons. Start at LOW power — you can always turn it up.

45-70 min

- ☐ Test with real Pollen. Find the power level that grabs reliably without jamming or launching it across the room. Write that number down.

70-90 min

- ☐ Add telemetry showing the intake state and power. If the motor stalls when jammed, add a guard that cuts power.

90-100 min

- ☐ Draw a gamepad control map — a picture of the controller with every used button labeled. Tape it to the Driver Station.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Intake picks up Pollen reliably on a button press
- ☐ Working power level recorded in the notebook
- ☐ Control map drawn and taped to the Driver Station

WHAT YOU NEED

- Built intake (BA-5)
- Configured intake motor (PR-2)
- Real Pollen to test with
- Paper for the control map

BEFORE YOU START

PR-2, BA-5

THIS UNLOCKS

IF YOU FINISH EARLY

- Make it hold-to-intake, release-to-hold, so the driver never has to think about turning it off.

HAND OFF TO

Show the drivers the control map and let them try it before you call it done.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

There is one robot and more than one person who wants to program. The browser simulators let everyone write and test code at the same time, with no waiting and nothing to break.

YOU WILL FINISH WITH

Everyone in the crew has driven and programmed a virtual robot through a course.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Open ftcsim.org (and vrobotsim.online as a second option). Get everyone onto a machine.

15-40 min

- ☐ Work through the intro tutorial. Drive the virtual robot manually first so you know what good driving looks like.

40-75 min

- ☐ Program a simple autonomous in the simulator: forward, turn, forward, stop. Same discipline as the real robot — one change at a time.

75-95 min

- ☐ Race each other. Whose simulated autonomous gets furthest through the course?

95-100 min

- ☐ Write down one thing the simulator taught you that you want to try on the real robot.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- One laptop or Chromebook per person
- ftcsim.org and/or vrobotsim.online
- No robot required

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

IF YOU FINISH EARLY

- Rebuild your real Wayfinder path in the simulator and compare how it behaves against the real thing.

DONE WHEN – ALL 2 MUST BE TRUE

- Everyone has a working simulated autonomous
- One transferable idea written down for the real robot

HAND OFF TO

This is the best task to run in parallel when the robot is tied up in a build task.

Who worked on this: _____ Date: _____

PR-10

Touch Sensor: The First Decision

WHY THIS MATTERS

Every interesting program makes a decision. A touch sensor is the simplest one there is — pressed or not pressed. Once you can write 'if this, then that', you can write an arm that stops at the top instead of tearing itself apart.

YOU WILL FINISH WITH

A program where pressing the touch sensor changes what the hardware does, with the sensor state live on telemetry.

WHAT YOU NEED

- Assembled programming board (BA-9)
- Touch sensor in the configuration (PR-2)
- FTC Docs digital sensor pages

BEFORE YOU START

PR-2

THIS UNLOCKS

PR-13

IF YOU FINISH EARLY

- Add a counter that displays how many times the sensor has been pressed. Then work out why one push sometimes counts as several, and fix it. The word to look up is 'debounce'.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm the touch sensor is in the configuration under a clear name like touch_sensor.
- ☐ Read the FTC Docs page on digital sensors.

15-40 min

- ☐ Write a program that does NOTHING but print the sensor state to telemetry. Press it, release it, watch the value flip.
- ☐ Do not move on until the number on screen matches exactly what your finger is doing.

40-70 min

- ☐ Add the decision: if the sensor is pressed, run the motor; if it is not, stop the motor. Test it.
- ☐ Now invert it, so the motor runs only when NOT pressed. Doing both is how you learn which is which.

70-90 min

- ☐ Make it useful. Use the touch sensor as a limit: the motor may run one direction freely, but stops the instant the sensor is pressed.
- ☐ This is exactly how a real arm or lift protects itself from driving into its own frame.

90-100 min

- ☐ Write in the notebook, in your own words, what if/else means, and name one place on a competition robot where you would put a limit switch.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Telemetry shows the sensor state changing correctly
- ☐ Motor behavior changes when the sensor is pressed
- ☐ Limit behavior stops the motor in one direction only
- ☐ Explanation written in the notebook

HAND OFF TO

PR-13 moves this logic onto the real robot.

Who worked on this: _____ Date: _____

Color Sensor: Teach It to See Pollen

WHY THIS MATTERS

This season's scoring element is Pollen. A robot that can tell whether it is actually holding one can make decisions the driver would otherwise have to make under pressure, while looking at the wrong thing.

YOU WILL FINISH WITH

A program that reliably reports whether a Pollen is in front of the sensor, using thresholds you measured rather than guessed.

WHAT YOU NEED

- Assembled programming board (BA-9)
- Color sensor in the configuration
- Real Pollen
- 3-4 other objects to test against
- FTC Docs color sensor pages
- Paper for your data table

BEFORE YOU START

PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Lighting changes everything for a color sensor. Test yours next to a window and again under only the room lights, and see how far your numbers move. That is why teams shroud their sensors.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm the color sensor is in the configuration. Read the FTC Docs page on the color sensor.

15-45 min

- ☐ Print the raw values to telemetry: red, green, blue, and the proximity reading.
- ☐ Hold a Pollen in front of it. Then your hand, the table, a sheet of white paper. WRITE DOWN the numbers for each one.
- ☐ You are collecting data here, not guessing. The table you build is the whole point of this hour.

45-70 min

- ☐ Look at your table. Pick a threshold that separates 'a Pollen is there' from 'nothing is there'.
- ☐ Write the rule as an English sentence BEFORE you write it as code.

70-90 min

- ☐ Code the rule. Test it 10 times with a Pollen and 10 times without, counting how many times it was right.
- ☐ Under 18 correct out of 20 means your threshold is wrong — go back to your numbers, do not just nudge the code.

90-100 min

- ☐ Record the final thresholds and the 20-trial result in the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ A written table of measured values for at least 4 different objects
- ☐ A threshold rule written in plain English before it was written in code
- ☐ 18 or more correct out of 20 trials
- ☐ Thresholds and results recorded

HAND OFF TO

Note that these thresholds are only valid for THIS sensor at THIS distance. Say so next to the numbers.

Who worked on this: _____ Date: _____

Potentiometer & Servo: Analog In, Position Out

WHY THIS MATTERS

A motor spins. A servo goes to a position and holds it. Confusing the two is the most common beginner mistake in FTC, and wiring a knob to a servo makes the difference obvious in about ten seconds.

YOU WILL FINISH WITH

Turning the potentiometer knob sweeps the servo to a matching position, with both values on screen.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Confirm both are in the configuration. Read the FTC Docs pages on analog input and on servos.

15-40 min

- ☐ Print the potentiometer's raw value to telemetry. Turn the knob slowly from one stop to the other and write down the value at each end.
- ☐ That pair of numbers is your input range. Everything else today depends on it.

40-60 min

- ☐ Separately, command the servo to fixed positions (0.0, 0.5, 1.0) on button presses. Watch it move there and HOLD.
- ☐ Compare that to the motor, which only spins. Say the difference out loud to your partner.

60-85 min

- ☐ Connect them: map the potentiometer's measured range onto the servo's 0 to 1 range, so turning the knob sweeps the servo. Show both the raw and the mapped value on telemetry.

85-100 min

- ☐ Write in the notebook, in your own words, the difference between a motor and a servo and where each one belongs on a competition robot.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Assembled programming board (BA-9)
- Potentiometer and servo in the configuration
- FTC Docs analog input and servo pages

BEFORE YOU START

PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add limits so the servo can never be commanded past a safe range even if the knob turns further. Real mechanisms break when code forgets this.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Knob sweeps the servo smoothly across its full travel
- ☐ Both raw and mapped values shown on telemetry
- ☐ Motor versus servo explained in writing

HAND OFF TO

This mapping technique works for any analog sensor. PR-13 moves it to the robot.

Who worked on this: _____ Date: _____

Bench to Robot: Move Your Code Over

WHY THIS MATTERS

Code that works on the bench does not automatically work on the robot — different ports, different names, different mechanical reality. Making that jump is a real engineering skill, and it is where the two halves of your team finally meet.

YOU WILL FINISH WITH

One behavior developed on the programming board, running on the actual robot.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pick ONE behavior to move over — not all of them.
- ☐ Put the bench port map and the robot port map next to each other and find every name and port that differs.
- ☐ If you are using the class robot, write down WHICH robot you configured — a config saved to one hub does not exist on the other.

15-40 min

- ☐ Mount the sensor on the robot somewhere it can actually do its job, and wire it.
- ☐ Add it to the ROBOT's configuration using the same name you used on the bench if you can. That alone removes a whole category of bugs.

40-70 min

- ☐ Copy the behavior into the robot's TeleOp. Expect it not to work the first time — that is normal, not failure.
- ☐ Use telemetry to find out whether the sensor is reading at all BEFORE you start blaming the logic.

70-90 min

- ☐ Test it on the robot in a realistic situation, not just sitting on the bench. Sensor readings will be different out there — retune the thresholds against the real thing.

90-100 min

- ☐ Write down what had to change between bench and robot, and why.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ The behavior runs on the actual robot
- ☐ Telemetry confirms the sensor reads correctly where it is mounted
- ☐ A written list of what changed from bench to robot

WHAT YOU NEED

- A working bench program (PR-10, PR-11 or PR-12)
- A robot with working TeleOp (PR-3) — either 22257's robot or the Herobotix class robot; both run a REV Control Hub
- Both port maps side by side
- A sensor you can spare to mount on the robot
- Hex drivers

BEFORE YOU START

PR-3, PR-10

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Teach the change back to whoever wrote the original bench version. Explaining it is how you find out whether you actually understood it.

HAND OFF TO

This is the pattern for the rest of the season: prototype on the bench, prove it on the robot.

Who worked on this: _____ Date: _____

Field-Centric Drive with the IMU

WHY THIS MATTERS

In normal driving, 'forward' means the robot's forward — so the moment the robot faces the driver, left and right swap and everyone gets lost. Field-centric drive uses the Control Hub's built-in IMU so 'forward' always means away from the driver. On a mecanum robot it is the single biggest driving upgrade available.

YOU WILL FINISH WITH

A drive mode where pushing the stick away from you moves the robot away from you, whichever way it is facing.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the FTC Docs pages on the Control Hub's built-in IMU.
- ☐ Print the robot's heading to telemetry and rotate the robot by hand while watching it. Confirm it reads about 0 when the robot faces your chosen 'forward'.

20-45 min

- ☐ Understand the idea before you code it: you rotate the driver's stick input by the **NEGATIVE** of the robot's current heading, then feed that result into the same mecanum mixing you wrote in PR-3.
- ☐ Draw it on paper with the robot turned 90 degrees before you touch the keyboard.

45-80 min

- ☐ Implement it. Add a button that resets the heading to zero — the driver needs it at the start of every match and you will need it constantly while testing.

80-95 min

- ☐ Test properly: point the robot in four different directions and each time push the stick straight away from you. It should move away from you every single time.

95-100 min

- ☐ Record what changed and let two different drivers try both modes.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Heading displays on telemetry and behaves correctly
- ☐ Reset-heading button works
- ☐ Robot moves away from the driver from all four starting orientations

WHAT YOU NEED

- Working mecanum TeleOp (PR-3)
- Telemetry (PR-4)
- Taped field (EP-6)
- FTC Docs IMU pages
- Two drivers to test with

BEFORE YOU START

PR-3, PR-4

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Put robot-centric and field-centric on a toggle button and let the drivers pick which they want for Fast Lane. Record which they chose and why.

HAND OFF TO

This is the mode most competitive teams drive with. Note which drivers prefer which mode.

Who worked on this: _____ Date: _____

Inventory the Kit & Set Up the Shop

No hardware

Start here

3-5 people

100 min

WHY THIS MATTERS

You cannot build with parts you cannot find. This is the highest-value hour a brand-new team ever spends, and it pays back every single meeting afterward.

YOU WILL FINISH WITH

Everything counted, labeled, and in a known place, with a written inventory and a photo of what clean looks like.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Clear a big table. Open the REV kit and pull out any existing team bins.

15-50 min

- ☐ SORT FIRST, count later. Group by type: structure and channel, motors, wheels, electronics, fasteners, tools, miscellaneous.

50-75 min

- ☐ Now count. Record into the inventory sheet: part name, quantity, and where it lives.

75-90 min

- ☐ Label every bin and drawer. Then take a photo of the shelf looking exactly the way it should look.

90-100 min

- ☐ Print the photo and post it above the shelf. That photo is now the team's cleanup standard — the last 10 minutes of every meeting is matching it.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The REV kit and every existing parts bin
- Bins, drawers, or containers
- Label tape or a label maker
- Laptop or paper for the inventory sheet

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

BA-2

IF YOU FINISH EARLY

- Flag anything you are low on and add it to the team's parts wish list for the coach.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Inventory sheet filled in with quantities and locations
- ☐ Every bin labeled
- ☐ Reference photo printed and posted above the shelf

HAND OFF TO

Show the whole team where things live. Point at the reference photo during the first cleanup.

Who worked on this: _____ Date: _____

Mecanum Chassis: Build the Frame

WHY THIS MATTERS

The drivetrain is the foundation — every other mechanism bolts to it. On a mecanum robot a frame that is not square means four wheels that fight each other instead of working together.

YOU WILL FINISH WITH

A square C-channel chassis built to the REV Mecanum Drivetrain Kit v2 build guide.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Open the Mecanum Drivetrain V2 build guide and read the **WHOLE** frame section before anyone picks up a tool.
- ☐ The guide splits the build into sections so people can work in parallel — decide now who takes which section.

15-30 min

- ☐ Lay out the C-channels and the brackets and fasteners the frame section calls for.
- ☐ **SORT THE CHANNELS BY LENGTH FIRST.** The kit has two lengths (408mm and 248mm) and mixing them up costs you an hour.

30-75 min

- ☐ Assemble the frame following the guide. Start every fastener by hand, then snug it — do **NOT** fully tighten until the frame is square.

75-90 min

- ☐ Check square: measure both diagonals of the frame. Equal diagonals means square. Adjust until they match, **THEN** tighten everything down.

90-100 min

- ☐ Photograph the finished frame into the notebook with the date and who built it.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- REV Mecanum Drivetrain Kit v2 (REV-45-2470)
- The Mecanum Drivetrain V2 build guide — link on the resource shelf
- Hex drivers (included in the kit)
- Tape measure
- Safety glasses

BEFORE YOU START

BA-1, EP-7

THIS UNLOCKS

BA-3

IF YOU FINISH EARLY

- Start a torque-check habit: at the start of every meeting, someone re-checks every fastener on the robot.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Diagonals measured within about 1/8 in of each other
- ☐ All fasteners tightened after squaring, not before
- ☐ Channel lengths match the guide, no unexplained leftover parts

HAND OFF TO

Hand off to BA-3 for gearboxes, motors and wheels.

Who worked on this: _____ Date: _____

Mecanum Drive: Gearboxes, Motors & Wheels

Robot

Some experience

3-4 people

100 min

WHY THIS MATTERS

This is where a frame becomes a robot that can move in any direction. It is also where mecanum goes wrong: the four wheels are NOT interchangeable, and one wheel in the wrong corner turns strafing into spinning.

YOU WILL FINISH WITH

Four motor and gearbox assemblies mounted with the mecanum wheels in the correct pattern, every wheel spinning freely.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Read the gearbox and wheel sections of the build guide.
- ☐ Each corner is an HD Hex motor with an UltraPlanetary gearbox plus an Ultra 90 to turn the output 90 degrees. Build ONE complete assembly together as a group first, so everybody has seen how it goes together, then split up.

20-60 min

- ☐ Build the remaining three assemblies and mount all four to the frame per the guide. Keep the left and right sides correctly mirrored.

60-80 min

- ☐ MECANUM WHEEL CHECK — this is the mistake every new team makes. The four wheels are handed and they are not interchangeable.
- ☐ Mount them, then look straight down at the robot from above. The top rollers should form an X across the chassis.
- ☐ If you see rollers running parallel, or a diamond instead of an X, two wheels are swapped. Check against the build guide photos or the Onshape CAD before you accept it.

80-95 min

- ☐ Spin every wheel by hand — anything that binds or rubs gets fixed NOW, not later.
- ☐ Set the robot on a flat floor and confirm all four wheels touch at the same time.

95-100 min

- ☐ Write down which motor is in which corner. The programming crew needs this for PR-2.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Squared frame (BA-2)
- Mecanum Drivetrain V2 build guide
- 4x UltraPlanetary gearbox kits with HD Hex motors
- 4x Ultra 90 degree gearboxes
- The 75mm mecanum wheel set
- Hex drivers
- A flat floor

BEFORE YOU START

BA-2

THIS UNLOCKS

BA-4, BA-5

IF YOU FINISH EARLY

- Push the robot sideways by hand across the floor. If the wheels are right it slides sideways smoothly. If it fights you or twists, re-check the X pattern before you blame the code later.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Four motor and gearbox assemblies mounted
- ☐ Rollers form an X viewed from above, verified against the guide
- ☐ Every wheel spins freely and the robot sits flat
- ☐ Corner positions written down for the programmers

HAND OFF TO

Give the corner positions to whoever does PR-2. If the X pattern is wrong, strafing will never work no matter how good the code is.

Who worked on this: _____ Date: _____

Electronics: Mount & Wire the Control Hub

Robot

Some experience

2-3 people

100 min

WHY THIS MATTERS

The Mecanum Drivetrain Kit includes no control system, no battery and no wiring — all of that is on you. Bad wiring is the number one reason a robot dies mid-match, and on a four-motor mecanum drive a single mislabeled cable turns strafing into spinning.

YOU WILL FINISH WITH

Control Hub, battery and switch mounted and wired, all four drive motors labeled at both ends and tied down.

WHAT YOU NEED

- Robot with drivetrain (BA-3)
- A REV Control Hub — NOT included in the kit; confirm with the coach that one is free before you start
- 12V battery and main power switch
- Motor cables and power cabling
- Zip ties
- Label tape
- Coach or mentor

BEFORE YOU START

BA-3, EP-7

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Make a laminated port map card that lives on the robot cart, so a mid-competition repair does not need a laptop.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Confirm with the coach that a Control Hub is actually available for this robot. The kit does not include one and the team's hubs are already spoken for — do not start until this is settled.
- ☐ Read the wiring guidance in the build guide. ASK THE COACH before cutting, crimping or modifying any cable.

20-55 min

- ☐ Mount the Control Hub where you can reach the ports AND see the status lights. Mount the battery so it physically cannot fall out or slide, and put the switch somewhere a referee can reach it.

55-80 min

- ☐ Route all four motor cables to the hub. Label BOTH ENDS of every cable: LF, RF, LB, RB.
- ☐ On a mecanum drive the corner labels ARE the robot's behavior — get them wrong and the strafe math in PR-3 will be chasing a wiring bug.

80-95 min

- ☐ Tie down every cable so nothing can reach a wheel or a moving part. Tug-test each connector gently to confirm it is seated.

95-100 min

- ☐ Photograph the wiring and write the port map into the notebook.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Control Hub, battery and switch mounted, battery cannot fall out
- ☐ All four motor cables labeled by corner at both ends
- ☐ Nothing loose, hub status lights visible, port map recorded

HAND OFF TO

The port map goes straight to PR-2. Any wiring change means the configuration changes too.

Who worked on this: _____ Date: _____

Design & Build an Intake

WHY THIS MATTERS

The Mecanum Drivetrain Kit is a drive base and nothing else — it ships with no intake and there are no instructions to follow. Whatever picks up Pollen on this robot, your team designs and builds. That is real FTC, and it is the part judges ask about.

YOU WILL FINISH WITH

A mounted intake prototype that picks up a Pollen, with a measured success rate.

WHAT YOU NEED

- Robot with drivetrain (BA-3)
- The concept chosen in EP-4
- Parts from the team's stock
- Cardboard and zip ties for mock-ups
- Real Pollen (10+)
- Hex drivers

BEFORE YOU START

BA-3, EP-4

THIS UNLOCKS

BA-7, PR-8

IF YOU FINISH EARLY

- Test a second mounting height, 10 more attempts, and record which grabbed better. One number beats an hour of arguing.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Review the concept chosen in EP-4 and gather parts. Check the Competition Manual sizing rule — whatever you add has to keep the robot legal at the start of a match.

15-45 min

- ☐ Build the MOUNTING first — how does this attach to the C-channel? Get that solid before you worry about the grabbing part. A great intake that falls off is worth nothing.

45-75 min

- ☐ Build the grabbing mechanism. Rough is fine at this stage; cardboard and zip ties are legitimate. Test it by hand with a Pollen as you go.

75-95 min

- ☐ Power it if you have a spare motor or servo. Do 10 pickup attempts and count how many succeed. Write the number down.

95-100 min

- ☐ Photograph it and record the mounting height and the success count — you will change the height later and need to know where you started.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Intake mounted solidly to the chassis
- ☐ Robot still fits the legal starting size with it attached
- ☐ 10 pickup attempts counted and the result recorded

HAND OFF TO

Tell the programming crew the intake exists — PR-8 can map it to a button.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Half of all 'the robot is acting weird' problems are a low battery. A team that manages batteries wins matches that other teams lose for no visible reason.

YOU WILL FINISH WITH

A labeled, color-coded charging station and a written battery routine everyone follows.

WHAT YOU NEED

- Every battery the team owns
- Chargers
- Battery status stickers (ask the coach — the team has printable sheets)
- Multimeter or battery checker
- Permanent marker

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Start a weekly voltage log. You will spot a dying battery weeks before it costs you a match.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Gather every battery in the room. Ask the coach for the team's battery sticker sheets.

15-40 min

- ☐ Number every battery and write the date it entered service on it. Apply a status sticker: charged / in use / needs charging / retired.

40-65 min

- ☐ Set up the station in ONE place: chargers labeled, a clearly marked shelf for charged batteries and a separate one for dead batteries.

65-85 min

- ☐ Measure and record each battery's resting voltage. Flag any battery that reads consistently low — it may be near the end of its life.

85-100 min

- ☐ Write the battery rules on a card and post it: who swaps, when, where dead ones go, and who is responsible for putting them on the charger.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Every battery numbered, dated, and stickered
- ☐ Resting voltages logged
- ☐ Station set up with charged/dead separated
- ☐ Rules card posted

HAND OFF TO

Announce the battery rules to the whole team. This only works if everyone follows it.

Who worked on this: _____ Date: _____

Scoring Mechanism Prototype (cardboard first)

Robot

Challenge

3-4 people

100 min

WHY THIS MATTERS

Getting Pollen in is one problem; putting it where you want it is a different one. Prototyping in cardboard is far faster and cheaper than discovering the flaw in aluminum.

YOU WILL FINISH WITH

A working rough prototype tested with real Pollen, with two versions and a success count for each.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Review the concept chosen in EP-4. Gather cardboard and scrap. Do not raid the good parts bin.

15-50 min

- ☐ Build the roughest version that could possibly work. Do NOT make it pretty. Speed matters far more than finish at this stage.

50-75 min

- ☐ Test with real Pollen. Do 10 attempts and count how many land where you wanted. Write the number down.

75-95 min

- ☐ Change exactly ONE thing. Test 10 more attempts. Record that number too.

95-100 min

- ☐ Photograph both versions and write down what you learned, including what surprised you.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The chosen concept from EP-4
- Cardboard, zip ties, tape, scrap parts
- Real Pollen (10+)
- Scissors or box cutter (with supervision)

BEFORE YOU START

BA-5, EP-4

THIS UNLOCKS

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IF YOU FINISH EARLY

- Sketch what the real metal version would look like and list every part you would need to order.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ A physical prototype exists
- ☐ Two versions tested with 10 attempts each and both counts recorded
- ☐ Findings written in the notebook

HAND OFF TO

Bring the numbers to the next planning session — they decide whether this concept is worth building for real.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

At a competition you get a small pit and very little time between matches. A team with a cart and an organized toolkit fixes the robot while other teams are still looking for a hex driver.

YOU WILL FINISH WITH

A cart the robot rides on safely and a packed pit toolkit with a checklist taped inside the lid.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-20 min**

- ☐ Find or borrow a cart. Set the robot on it and check that it is stable and will not slide off when the cart is pushed.

20-50 min

- ☐ Pack the pit toolkit: hex drivers, spare fasteners, spare motor cables, zip ties, tape, a charged spare battery, and a multimeter if the team has one.

50-75 min

- ☐ Write the pit checklist — one line per item that must be in the box. Print it and tape it inside the lid.

75-95 min

- ☐ Dry run: unpack everything, repack it, and have a DIFFERENT person check it against the list.

95-100 min

- ☐ Park the cart where it will live. Photograph the packed kit.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Robot rides on the cart without sliding
- ☐ Toolkit packed
- ☐ Checklist taped inside the lid and verified by a second person

HAND OFF TO

The pit kit gets checked against its list before every event. Name who owns that.

Who worked on this: _____ Date: _____

WHAT YOU NEED

- A cart or rolling cabinet (ask the coach)
- A toolbox or bin
- Tools and spares
- Label tape
- Printer

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Add a laminated 5-minute repair card listing the three most likely failures and how to fix each one.

Assemble a Programming Board

WHY THIS MATTERS

22257's first board is already built — this card is for building a SECOND one, or repairing the first. The board carries a Control Hub, a motor, a servo and three sensors, so a programming crew can work with none of the robot.

YOU WILL FINISH WITH

A fully assembled, wired programming board that powers up, with its port map taped to it.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Check with the coach first: is there a spare Control Hub for this board? Without one the board cannot run, and the existing board already has the team's second hub.
- ☐ Lay out every printed part and every component and check them against the model's parts list. One missing M4x40 stops the whole build.
- ☐ This is the REV build — use the REV cover and REV motor mounts, NOT the goBILDA ones. They are not interchangeable.

15-45 min

- ☐ Mount the Control Hub (4x M3x18mm with M3 lock nuts) and the potentiometer (4x M3x6mm).
- ☐ These threads go into printed plastic. Snug them and STOP — overtightening strips the part and you will have to reprint it.

45-70 min

- ☐ Mount the servo (4x M4x40mm), the touch sensor (2x M4x12mm) and the color sensor (2x M4x12mm).
- ☐ Mount the motor: front mount (2x M4x14mm to the board, 4x M3x8mm to the motor), then the rear support and clamp (2x M4x40mm).
- ☐ Fit the motor wheel and press on the potentiometer knob.

70-90 min

- ☐ Wire it: motor to a motor port, servo to a servo port, touch sensor and potentiometer to their ports, color sensor to an I2C port.
- ☐ WRITE DOWN which port every single thing went to. The next crew needs this list for PR-2 and cannot guess it.

90-100 min

- ☐ Power it from a charged 12V battery and confirm the Control Hub lights come up. Tape the port map to the board.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Every component mounted, nothing loose, no stripped printed parts
- ☐ Port map written and taped to the board
- ☐ Hub powers up on battery

WHAT YOU NEED

- A spare REV Control Hub — confirm with the coach BEFORE starting; the existing board already has the team's second hub
- Printed parts, REV variant: board, cover, front motor mount, rear support, rear clamp, motor wheel, potentiometer knob
- REV components: HD Hex Motor 40:1 (41-1301), Smart Robot Servo (41-1097), Touch Sensor (41-1425), Color Sensor V3 (31-1557), Potentiometer (31-1155)
- M3 and M4 hardware plus nylon lock nuts — see the model's parts list
- Hex drivers and a charged 12V battery

BEFORE YOU START

EP-7

THIS UNLOCKS

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IF YOU FINISH EARLY

- If the team has a second Control Hub and a second set of printed parts, build a second board. Two boards means two programming crews at once.

HAND OFF TO

Give the port map to whoever takes PR-2. Cards PR-10, PR-11 and PR-12 all need this board.

Who worked on this: _____ Date: _____

MG-1**MERCH & GIVE-AWAY****Give-Away Plan & Budget**

No hardware

Start here

2-3 people

100 min

WHY THIS MATTERS

Trading give-aways with other teams is one of the best parts of a competition and it is how people remember you afterward. But it costs real money, so decide before you buy.

YOU WILL FINISH WITH

One chosen give-away with a real cost per unit, a justified quantity, and a total that fits the budget.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Get the give-away budget from the coach. Write it on the board where everyone can see it.

15-40 min

- ☐ Brainstorm at least 8 options: buttons and pins, stickers, 3D-printed tokens, keychains, trading cards, bracelets, patches, magnets.

40-65 min

- ☐ For your top 3, find REAL prices online, or calculate material cost if you are making them. Work out cost per unit and total.

65-85 min

- ☐ Justify the quantity: number of teams at your events multiplied by 2, plus extras for judges, sponsors, and volunteers. Show the math on the board.

85-100 min

- ☐ Pick one primary and one backup. Write a one-paragraph recommendation to the coach with the numbers in it.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- The give-away budget number (ask the coach)
- Laptop for pricing
- Whiteboard
- Calculator

BEFORE YOU START

EP-1

THIS UNLOCKS

MG-2, MG-5

IF YOU FINISH EARLY

- Price a second, nicer item to hand specifically to sponsors and judges.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ One give-away chosen with a real cost per unit
- ☐ Quantity justified with visible math
- ☐ Total is under budget
- ☐ Written recommendation handed to the coach

HAND OFF TO

The coach approves the spend. Once approved, MG-2 designs the art.

Who worked on this: _____ Date: _____

MG-2

MERCH & GIVE-AWAY

Design the Give-Away Art

No hardware

Some experience

2-4 people

100 min

WHY THIS MATTERS

The art carries the team name to people who have never met you. It has to be readable from three feet away in about one second.

YOU WILL FINISH WITH

Print-ready artwork at the correct size and resolution, saved in the team's shared folder.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Pull up the identity brief and logo sketches from EP-1. Get the exact dimensions of the item from MG-1.

15-50 min

- ☐ Draft 3 versions. Keep it simple — a button is small and a sticker gets looked at for one second. Team name AND team number 22257 must both appear.

50-70 min

- ☐ The squint test: print a draft at actual size, stand 6 feet back. Can you still read the name and number? If not, make them bigger and cut something else.

70-90 min

- ☐ Show it to someone not on the crew. Ask what they noticed first. Revise once based on what they say.

90-100 min

- ☐ Export at the right physical size and 300 dpi for print. Save to the team shared folder with a clear filename including the date.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Passes the 6-foot squint test
- ☐ Team name and number 22257 both present and legible
- ☐ Exported at correct size and 300 dpi
- ☐ Saved in the shared folder with a dated filename

WHAT YOU NEED

- Identity brief and logo sketches (EP-1)
- The exact size and shape of the item (MG-1)
- Design software or Canva
- A printer for test prints

BEFORE YOU START

EP-1, MG-1

THIS UNLOCKS

MG-3

IF YOU FINISH EARLY

- Make a matching version sized for a sticker and another sized for a social media post.

HAND OFF TO

Give the file to MG-3 for production, and to the MF-4 social crew to reuse.

Who worked on this: _____ Date: _____

Produce the Give-Aways (assembly line)

WHY THIS MATTERS

Making 200 of something is a completely different job from making one. Run it like a production line and it is fun; run it randomly and it is miserable.

YOU WILL FINISH WITH

A counted, bagged, quality-checked batch stored with the competition gear.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Set up stations. Assign roles: cut, assemble, quality check, count and bag.

15-25 min

- ☐ Make ONE complete unit together. Everyone looks at it. Agree out loud that this is the standard.

25-80 min

- ☐ Run the line. Rotate stations halfway through so nobody is stuck on one job for an hour.

80-90 min

- ☐ Quality check: pull 10 at random and inspect them. If there are rejects, fix the PROCESS, not just those pieces.

90-100 min

- ☐ Count the final total, bag them, write the count on the bag, and store it with the pit kit.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Approved art (MG-2)
- Blanks and supplies
- Button press / cutter / printer as needed
- Zip bags
- Marker

BEFORE YOU START

MG-2

THIS UNLOCKS

IF YOU FINISH EARLY

- Record how many you produced per minute. You will want that number the next time you plan a batch.

DONE WHEN – ALL 3 MUST BE TRUE

- ☐ Batch counted and bagged with the count written on the bag
- ☐ Random sample of 10 passed inspection
- ☐ Stored with the competition gear

HAND OFF TO

Tell the coach the count. Add the bag to the pit kit checklist (BA-8).

Who worked on this: _____ Date: _____

MG-4**MERCH & GIVE-AWAY**

Team Card / Business Card

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

Judges, sponsors, and other teams all ask 'how do I find you?'. A card answers that in one hand-off and it stays in their pocket.

YOU WILL FINISH WITH

A proofread, print-ready card design with a known print source and cost.

WHAT YOU NEED

- Identity brief and logo (EP-1)
- Team photo
- Design software
- Printer for a test page
- Scissors

BEFORE YOU START

EP-1

THIS UNLOCKS

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IF YOU FINISH EARLY

- Add a QR code pointing at the team's page on the Herobotix website.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Decide the content: team number 22257, team name, school, season, a photo or logo, and ONE contact.
- ☐ IMPORTANT: use a team email or the coach's contact. Never a student's personal phone, email, or social handle.

15-45 min

- ☐ Lay out front and back at standard card size, 3.5 x 2 inches. Keep everything at least 1/8 inch inside the edge so nothing gets trimmed off.

45-65 min

- ☐ Proofread it. Then have two other people proofread it. A typo on 200 printed cards is permanent.

65-85 min

- ☐ Print a test page on plain paper, cut one out, and hold it in your hand. Fix whatever looks wrong at real size.

85-100 min

- ☐ Save the print-ready file and write down where you would print it and what it costs.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Proofread by 3 different people
- ☐ Test print cut out and checked at actual size
- ☐ No student personal contact info anywhere on it
- ☐ Print source and cost written down

HAND OFF TO

Coach approves before printing. Cards go in the pit kit for competitions.

Who worked on this: _____ Date: _____

MG-5**MERCH & GIVE-AWAY**

3D-Printed Give-Away

No hardware

Challenge

2-3 people

100 min

WHY THIS MATTERS

A printed item is cheap per unit and unmistakably yours. It is also real CAD practice that transfers directly to designing robot parts.

YOU WILL FINISH WITH

One test print in hand and a realistic run plan with a start date.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-15 min**

- ☐ Confirm printer access and which filament colors are available. Get the coach's OK on how much filament you may use.

15-50 min

- ☐ Model or modify a simple item in Onshape: a keychain, badge, or Pollen-themed token. Emboss 22257 into it so it is unmistakably yours.

50-70 min

- ☐ Slice it and read the estimated print time. Multiply by how many you need. Is that actually possible before the event? If not, shrink it or simplify it until it is.

70-90 min

- ☐ Print ONE. Inspect it: does the number read clearly? Does it survive being dropped on the floor?

95-100 min

- ☐ Write the run plan: units per plate, hours per plate, plates needed, and the date printing must START.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- 3D printer access (confirm with the coach)
- Filament
- Onshape or slicer software
- Calipers or ruler

BEFORE YOU START

MG-1

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Design a version that snaps onto a lanyard or a backpack zipper — people keep those.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ One test print in hand and inspected
- ☐ Print time math done and shown
- ☐ Run plan written with a start date

HAND OFF TO

Give the start date to the coach. If it is soon, start printing.

Who worked on this: _____ Date: _____

Design the Award Patch Set

No hardware

Some experience

2-4 people

100 min

WHY THIS MATTERS

The team earns patches for finishing cards, and they go on the team shirt. Designing them yourselves beats buying generic ones — and a set that shares one frame looks like a set instead of seven unrelated stickers.

YOU WILL FINISH WITH

One shared patch frame plus seven category marks, drawn to rules that work as a heat transfer, an embroidered patch, or a pin.

WHAT YOU NEED

- Identity brief and logo from EP-1
- The team awards sheet, so you know how many designs are needed
- Paper, pencils and a ruler
- Design software
- A printer, to check at actual size
- A shirt to hold them against

BEFORE YOU START

EP-1

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Design the three special awards too — Assist, First Light and Keystone — so the whole set is finished and the coach can quote it in one go.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the awards sheet and count exactly what you have to design: one shared frame, seven category marks, and optionally the three special awards.
- ☐ Get the team logo and colors from EP-1. These patches have to look like they belong to this team.

15-35 min

- ☐ Design the FRAME first — the shape every patch shares. Keep it under 2.5 inches wide and keep it simple: a circle, a shield, a hex.
- ☐ The team name and number live on the frame, so you only draw them once instead of seven times.

35-75 min

- ☐ Design the seven category marks that drop into the frame. One clear idea each, drawn thick. The mark has to read at two inches from across a room, so a whole scene will not work — one object will.

75-90 min

- ☐ THE RULES CHECK. Nobody has decided yet whether these get pressed, embroidered or made into pins, so every design has to survive all three. Check all four rules on every design:
- ☐ No line thinner than 1/16 inch. No more than 4 colors. No small text. Still readable printed in black and white only.
- ☐ Anything that fails a rule gets simplified now, not after somebody has paid for it.

90-100 min

- ☐ Print them at ACTUAL SIZE, cut them out, and hold them on a shirt sleeve. Fix whatever does not work when it is really two inches wide.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ One shared frame designed, carrying the team name and number
- ☐ Seven category marks that drop into that frame
- ☐ All four artwork rules passed on every design
- ☐ Printed at actual size and checked on an actual sleeve

HAND OFF TO

Files go to the coach for quoting. Tell the TS-2 crew where on the shirt the patches will sit.

Who worked on this: _____ Date: _____

TS-1

TEAM T-SHIRTS

Shirt Design Concepts

No hardware

Start here

3-5 people

100 min

WHY THIS MATTERS

The shirt is what the team looks like to everyone at an event. Everybody should get a say before it is decided, or somebody spends the season in a shirt they hate.

YOU WILL FINISH WITH

Three or more concepts drawn front and back on shirt templates, presented, with written feedback collected.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Print or draw blank T-shirt outlines. Review the identity brief and logo sketches from EP-1.

15-25 min

- ☐ Get the rules from the coach and write them on the board: how many print colors are affordable, is the back allowed, must the school or Herobotix logo appear, does sponsor space need to be reserved on the back.

25-35 min

- ☐ RESERVE THE PATCH ZONE. This team earns award patches and they go on the shirt.
- ☐ Pick where they will sit — a sleeve or a panel on the upper back — and draw that area on every blank before you design anything.
- ☐ Nobody designs over it. Artwork under a patch is artwork nobody sees, and you cannot fix it after the shirts are printed.

35-70 min

- ☐ Split into small groups. Each group designs one complete concept, front and back, keeping the patch zone clear. Team number 22257 must be readable from across a gym.

70-90 min

- ☐ Post every concept. Each group presents theirs for 2 minutes.

90-100 min

- ☐ Collect written feedback on sticky notes stuck to each design. Do NOT vote today — that is TS-2, and sleeping on it produces better decisions.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ 3+ concepts drawn front and back
- ☐ Print-color and logo rules written down
- ☐ The patch zone drawn on every concept and kept clear
- ☐ Every concept presented
- ☐ Sticky-note feedback collected

WHAT YOU NEED

- Printed blank T-shirt outlines, front and back
- Colored markers
- Identity brief (EP-1)
- Sticky notes

BEFORE YOU START

EP-1

THIS UNLOCKS

TS-2

IF YOU FINISH EARLY

- Sketch how the winning direction would look on a hoodie or a hat.

HAND OFF TO

Leave the designs and feedback up on the wall for TS-2.

Who worked on this: _____ Date: _____

Vote & Make the Art Print-Ready

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

A printer needs a real file, not a marker sketch. This is where a drawing becomes something that can actually be manufactured.

YOU WILL FINISH WITH

One winning design as a clean digital file, coach-approved, saved with a version number.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Run the vote. Every member gets one vote. The coach holds a veto only on rule and budget issues. Record the result in the notebook.

20-60 min

- ☐ Rebuild the winner as clean digital art. Vector format (SVG or AI) if you possibly can — printers ask for it. Keep contrast high and colors limited.

60-80 min

- ☐ Check every requirement: number 22257 legible, correct colors, correct placement, sponsor space reserved on the back if that was promised.
- ☐ Confirm the patch zone from TS-1 is still clear in the final art, and mark it on the file you hand the printer so they do not fill it.

80-95 min

- ☐ Get coach approval. Make any required changes right now while everyone is here.

95-100 min

- ☐ Save the final file into the shared folder with a version number and the date in the filename.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Vote recorded
- ☐ Clean digital file exists, vector if possible
- ☐ Patch zone preserved and marked on the file
- ☐ Coach approved
- ☐ Saved with version number and date

WHAT YOU NEED

- TS-1 concepts and feedback
- Design software (vector preferred: Inkscape, Illustrator, or Canva)
- Laptop
- Coach available to approve

BEFORE YOU START

TS-1

THIS UNLOCKS

TS-4

IF YOU FINISH EARLY

- Export a one-color version too — some printing methods are dramatically cheaper that way.

HAND OFF TO

The approved file goes to TS-4 for quotes.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The single most common shirt disaster is wrong sizes. Collect them properly, in person, once. This can run in parallel while others design.

YOU WILL FINISH WITH

A complete size count covering every member, mentor, and any parents ordering, with spares added.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Build a simple form: name, role (student / mentor / parent), size, adult or youth, quantity, paid yes/no.

20-40 min

- ☐ Bring an actual sample shirt if possible. People guess wrong from a size chart and guess right when they hold one.

40-70 min

- ☐ Collect from everyone present today. Then list who is missing and message them through the team Band group.

70-90 min

- ☐ Total it into an order sheet: sizes across the top, quantities down. Add 2-3 spares in the most common sizes.

90-100 min

- ☐ Hand the totals to the coach along with a hard deadline for anyone who has not responded.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ Every present member has a size recorded
- ☐ Missing people contacted through Band
- ☐ Totals sheet complete with spares
- ☐ Deadline set and communicated

WHAT YOU NEED

- Paper form or a simple online form
- A sample shirt in a couple of sizes if you can get one
- Access to the team's Band group for stragglers

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

TS-4

IF YOU FINISH EARLY

- Track who has paid, so the treasurer is not chasing it in three weeks.

HAND OFF TO

TS-4 needs these quantities to get comparable quotes.

Who worked on this: _____ Date: _____

Get Quotes & Recommend a Printer

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

The same shirt can cost twice as much depending on where you order it. Comparing quotes properly is a real business skill and it is judged.

YOU WILL FINISH WITH

Three comparable written quotes in a table, plus a written recommendation with an order-by date.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-20 min

- ☐ Write down the exact spec: quantity by size, number of print colors, front and back, and the date you need them in hand. Every quote must be for the SAME spec or you cannot compare them.
- ☐ Get the coach's OK before contacting any business.

20-70 min

- ☐ Get 3 quotes: one local shop, one online printer, and the cost of doing heat transfers in-house (the team has done transfers before — ask the coach).

70-85 min

- ☐ Build a comparison table: cost per shirt, setup fees, shipping, turnaround in days, and shirt brand/quality.

85-100 min

- ☐ Write a recommendation with a reason. Work backward from your event date to an ORDER BY date and put it in bold.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Approved art (TS-2)
- Size totals (TS-3)
- Laptop and email
- Coach approval before contacting anyone

BEFORE YOU START

TS-2, TS-3

THIS UNLOCKS

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IF YOU FINISH EARLY

- Ask each vendor whether they discount for a school robotics team. Many do, and nobody ever asks.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ 3 quotes for identical specs, in a table
- ☐ Written recommendation with a stated reason
- ☐ Order-by date calculated from the event date

HAND OFF TO

The coach places the order. Give them the table, the recommendation, and the deadline.

Who worked on this: _____ Date: _____

Build the Sponsor Prospect List

No hardware

Start here

2-4 people

100 min

WHY THIS MATTERS

Fundraising is not asking strangers. It is asking the right people, and the right people are almost always connected to somebody already on the team.

YOU WILL FINISH WITH

25 or more real prospects with a reason and a personal connection noted for each, ranked A/B/C.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Get the fundraising goal from the coach and write it on the board. Everything today is measured against that number.

15-40 min

- ☐ Everyone lists out loud: where their parents work, businesses their family uses regularly, local engineering / tech / manufacturing / medical companies, and businesses within a mile of the school.

40-70 min

- ☐ Build the sheet with columns: business name, what they do, why they would care about robotics, who on the team has a connection, contact info if known.

70-90 min

- ☐ Rank each one A (a real personal connection), B (good fit but no connection), or C (long shot).

90-100 min

- ☐ Hand the list to the coach. **DO NOT CONTACT ANYONE YET** — the coach approves all outreach first.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Laptop or shared sheet
- Whiteboard
- The team's fundraising goal (ask the coach)

BEFORE YOU START

Nothing — you can start this today

THIS UNLOCKS

MF-2

IF YOU FINISH EARLY

- Check whether any of them publish a community-giving page, grant program, or employee matching program.

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ 25+ prospects listed
- ☐ Connection and reason columns filled for each
- ☐ Ranked A/B/C
- ☐ Handed to the coach with nobody contacted

HAND OFF TO

The coach decides who gets contacted and how. MF-2 builds the thing you send them.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

A sponsor needs to know in 30 seconds who you are, what their money does, and what they get back. If it takes longer than one page, it does not get read.

YOU WILL FINISH WITH

A one-page sponsorship document with specific, deliverable tiers, reviewed by the coach.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-20 min**

- ☐ Gather the facts: team number 22257, school, number of students, what FIRST Tech Challenge is, what season it is, and what a season actually costs. Get REAL numbers from the coach — made-up numbers are worse than no numbers.

20-50 min

- ☐ Draft the page in four blocks: who we are (3 sentences), what we do, what we need, what you get.

50-75 min

- ☐ Build the tiers. Be concrete at each dollar level: logo on the shirt, logo on the robot, name on the website, a team visit to your office, a framed thank-you.
- ☐ Do not promise anything the team cannot actually deliver. A broken promise to a sponsor ends the relationship.

75-90 min

- ☐ Add a photo and the team contact — team email or the coach. NEVER a student's personal contact.

90-100 min

- ☐ Coach reviews before this leaves the room.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Identity brief and logo (EP-1)
- Real season budget numbers (ask the coach)
- A good team photo
- Design software or a word processor

BEFORE YOU START

EP-1, MF-1

THIS UNLOCKS

MF-3, MF-7

IF YOU FINISH EARLY

- Make a matching half-page version to hand out at events and demos.

DONE WHEN — ALL 5 MUST BE TRUE

- ☐ Fits on one page
- ☐ Tiers are specific and genuinely deliverable
- ☐ Real budget numbers used
- ☐ No student personal contact info
- ☐ Coach reviewed

HAND OFF TO

MF-3 writes the emails that carry this. MF-7 reuses the language for grant applications.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

The ask is the hardest part. Writing it together as a team makes it better and takes most of the fear out of it.

YOU WILL FINISH WITH

Three coach-approved templates — cold email, warm introduction, and thank-you — plus a tracking sheet.

WORK THE LIST – CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Read the MF-2 one-pager so the emails match what it promises.

15-55 min

- ☐ Draft all three. Keep each UNDER 200 WORDS. Each one says: who we are, what we are asking for, what they get, and one clear next step.

55-75 min

- ☐ Read each one out loud. Anything that sounds stiff or fake, rewrite in normal words. Delete every sentence that is not doing work.

75-90 min

- ☐ Coach edits and approves. NOTHING gets sent by a student without coach approval — that is a hard team rule.

90-100 min

- ☐ Save the approved templates and build a tracking sheet: who, date contacted, response, follow-up date.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Sponsorship one-pager (MF-2)
- Laptop
- Coach available to approve

BEFORE YOU START

MF-2

THIS UNLOCKS

IF YOU FINISH EARLY

- Write the 30-second spoken version you would use if you walked into the business in person.

DONE WHEN – ALL 4 MUST BE TRUE

- ☐ 3 templates, each under 200 words
- ☐ Coach approved
- ☐ Tracking sheet created
- ☐ Nothing sent yet

HAND OFF TO

The coach decides who sends what and when. Keep the tracking sheet current.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Sponsors, judges, and future team members all look you up. A page with three good posts beats a page with nothing, and it beats an abandoned page with one post from October.

YOU WILL FINISH WITH

Written posting rules, three coach-approved posts ready to go, and a named owner.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-20 min**

- ☐ Confirm with the coach which accounts the team may use and write down the rules: no student full names or faces without permission on file, no personal contact info, coach approves every post before it goes up.

20-40 min

- ☐ Plan the rhythm: what gets posted (build progress, competition results, sponsor thank-yous, member spotlights) and how often. Pick something you can actually sustain.

40-80 min

- ☐ Draft 3 real posts — caption plus the specific photo. Take the photos today if you do not already have them.

80-95 min

- ☐ Coach reviews every caption AND every photo, including who is visible in each one.

95-100 min

- ☐ Schedule them or hand them off. Write down who owns posting each week.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Photo and privacy rules written down
- ☐ 3 posts drafted with photos attached
- ☐ Coach approved every post and photo
- ☐ Weekly owner named

WHAT YOU NEED

- Identity brief (EP-1)
- Camera or phone
- Coach available to approve
- Access to whichever accounts the team uses

BEFORE YOU START

EP-1

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Write the sponsor thank-you post template now, so it is ready the moment a sponsor says yes.

HAND OFF TO

Give the rules to everyone who takes photos at meetings and events.

Who worked on this: _____ Date: _____

Team Page Content for the Herobotix Website

No hardware

Some experience

2-3 people

100 min

WHY THIS MATTERS

The team needs a page on the Herobotix site, and whoever has website access can only publish what you hand them. Someone has to actually write it.

YOU WILL FINISH WITH

A finished content package — copy plus images plus instructions — in one folder, ready to hand off.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-15 min

- ☐ Look at the existing Herobotix website and the other teams' pages. Match their format so it drops in cleanly.

15-50 min

- ☐ Write the copy: team number and name, a two-paragraph 'who we are', what this season's game is, meeting times, how to join, how to sponsor.

50-75 min

- ☐ Gather the images. CHECK WITH THE COACH that every student pictured has photo permission on file.

75-90 min

- ☐ Proofread twice. Verify every fact: team number, dates, and the spelling of every name.

90-100 min

- ☐ Put the text and images in ONE clearly named shared folder with a short note saying what goes where.

NOTES, SKETCHES, WHAT SURPRISED YOU

WHAT YOU NEED

- Identity brief and logo (EP-1)
- Team photo and robot photo
- Laptop
- Photo permission list from the coach

BEFORE YOU START

EP-1

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Write a shorter version suitable for a school newsletter or a season kickoff announcement.

DONE WHEN — ALL 3 MUST BE TRUE

- ☐ Copy written and proofread twice
- ☐ Images collected with photo permissions verified
- ☐ Everything in one named folder with a placement note

HAND OFF TO

Hand the folder to the coach, who passes it to whoever administers the Herobotix site.

Who worked on this: _____ Date: _____

WHY THIS MATTERS

Outreach is judged at competition, it recruits next year's team, and it is genuinely fun. Elementary schools lose their minds over a robot.

YOU WILL FINISH WITH

A complete 20-minute run-of-show with a materials list and assigned roles, ready to hand to the coach for scheduling.

WORK THE LIST — CHECK EACH BOX AS YOU GO**0-20 min**

- ☐ Pick the audience and venue: elementary school, library, school open house, a sponsor's office. Check with the coach on what is actually schedulable.

20-50 min

- ☐ Design the 20 minutes: what you show, who says what, and what the audience gets to TOUCH or DRIVE. Make it hands-on — watching a robot is boring after 90 seconds.

50-70 min

- ☐ List everything you need: robot, charged batteries, charged spares, field elements, table, extension cord, give-aways, a team sign.

70-90 min

- ☐ Assign roles and write the run-of-show minute by minute, so nobody is standing around wondering what happens next.

90-100 min

- ☐ Write the request to the coach: proposed dates, what you need from adults, and how the robot gets there.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Minute-by-minute 20-minute run-of-show written
- ☐ Complete materials list
- ☐ Roles assigned by name
- ☐ Written request handed to the coach

WHAT YOU NEED

- A robot that drives (PR-3)
- Paper for planning
- Laptop
- Coach's input on what is realistic

BEFORE YOU START

PR-3

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Write down 5 questions a 3rd grader would actually ask and the answers, so nobody freezes.

HAND OFF TO

The coach schedules it. Rehearse once before the real thing.

Who worked on this: _____ Date: _____

Grants & Team Funding Applications

No hardware

Challenge

2 people

100 min

WHY THIS MATTERS

FIRST and its sponsors offer grants that most teams never apply for, because nobody read the page. This is money that rewards paperwork.

YOU WILL FINISH WITH

A table of every grant the team qualifies for with deadlines, and one application draft started.

WORK THE LIST — CHECK EACH BOX AS YOU GO

0-25 min

- ☐ With the coach, go through the FIRST Dashboard and FIRST's grant and funding resources.
- ☐ Also check local sources: the school PTO, city or community foundations, and employer matching programs at parents' companies.

25-50 min

- ☐ Build the table: grant name, amount, deadline, who is allowed to apply (student or adult), and what the application requires.

50-60 min

- ☐ Sort by deadline. Flag anything due within the next 60 days in red.

60-95 min

- ☐ Take the nearest deadline and draft answers to its questions. Reuse the language from the MF-2 one-pager rather than starting from scratch.

95-100 min

- ☐ Hand the table and the draft to the coach — most applications must be submitted by an adult.

NOTES, SKETCHES, WHAT SURPRISED YOU

DONE WHEN — ALL 4 MUST BE TRUE

- ☐ Grant table with amounts and deadlines
- ☐ Anything due within 60 days flagged
- ☐ One application draft started
- ☐ Handed to the coach

WHAT YOU NEED

- FIRST Dashboard access (with the coach)
- Sponsorship one-pager (MF-2)
- Laptop
- Coach working alongside you

BEFORE YOU START

MF-2

THIS UNLOCKS

—

IF YOU FINISH EARLY

- Put a calendar reminder two weeks before every deadline on the list.

HAND OFF TO

The coach submits. Put the deadlines on the team calendar.

Who worked on this: _____ Date: _____